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DATABASE MANAGEMENT SYSTEMS (DBMS)

PROJECT REPORT on

Cricket Player Performance & Selection Intelligence System

Submitted by:

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Abstract

The Cricket Player Performance & Selection Intelligence System is a database-driven web application for organizing and analyzing cricket statistics to support player evaluation and selection. MySQL stores structured information about players, teams, matches, venues, innings, and batting and bowling performances. MongoDB stores flexible scouting reports and observations. The Next.js application provides player views, selection analytics, reports, scouting workflows, and role-protected administration. The project demonstrates relational design and integrity constraints, SQL data operations and analytics, MongoDB document operations and aggregation, authentication, and database-backed application integration. The report presents the model, implementation evidence, query demonstrations, application structure, and evaluation criteria.

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1. Introduction

Problem Statement

Cricket selection committees need to compare players across formats, venues, opponents and playing conditions. Career totals alone can hide context-specific strengths and weaknesses. This project provides a centralized database-driven system to store cricket records and generate performance reports that support informed player evaluation.

Objectives and Scope

The project organizes player, team, match, venue, innings, batting, bowling, dismissal and role data; supports analysis across ODI and T20I matches; maintains scouting reports; and presents player information and analytical reports through a web application. It is a decision-support system, not a live-scoring or betting application.

• Manage player profiles, team and match information, venues and weather.

• Record batting and bowling performance and dismissal details.

• Analyze player performance by format, opponent, venue and match conditions.

• Demonstrate SQL operations and advanced database concepts alongside MongoDB document operations.

• Provide an authenticated web interface for player analytics and scouting workflows.

2. User Requirements Specification – Tech Stack

Functional Requirements

• Browse players and performance information.

• Display batting analytics, selection views and advanced SQL reports.

• Create, view, edit, search, filter and delete scouting reports.

• Authenticate users and enforce administrator and analyst access rules.

• Validate input and maintain referential and domain integrity.

Frontend Technologies

Next.js, React and TypeScript provide application pages, reusable interface components and typed client code. Tailwind CSS is used for interface styling.

Backend Technologies
Next.js API routes handle application requests, validation and database operations. Node.js runs server-side code. The mysql2/promise and MongoDB Node.js drivers connect to the two databases. JWT supports sessions and bcrypt hashes passwords.

Database Technologies
MySQL stores structured cricket and account data in cricket_selection. MongoDB stores flexible scouting-report documents in cricket_selection_nosql, principally in the scouting_reports collection.

Development Tools
The project materials identify Visual Studio Code, MySQL Workbench, mongosh, Git and GitHub. MongoDB is also described as running locally or through Docker.

Justification of Technology Choices
The relational model fits the linked player, match, innings and performance records and supports keys, constraints, joins and analytical SQL. A Next.js/React interface provides the web views while its API routes connect the interface to the databases. TypeScript supports typed application code; JWT and bcrypt are used for session handling and password hashing.

Justification for MongoDB
Scouting observations can vary between reports, so document storage offers a flexible structure. MongoDB complements MySQL by managing scouting reports while MySQL remains the source for structured player and match records. The application checks player existence in MySQL before scouting-report writes. Vector search is not evidenced as implemented in the supplied project materials; it is described as a possible future enhancement.

3. E-R Model

List of Entities
The conceptual model contains the following entities: Player, Team, Match, Venue, Weather, Innings, Player Role, Batting Performance, Bowling Performance, Dismissal, Opponent and Users. The supplied ER model does not identify any weak entity.

List of Non-Simple Attributes
The supplied ER model represents the listed entity attributes as individual scalar fields. No composite or multivalued attributes are specified in the model.

ER Diagram Snapshot

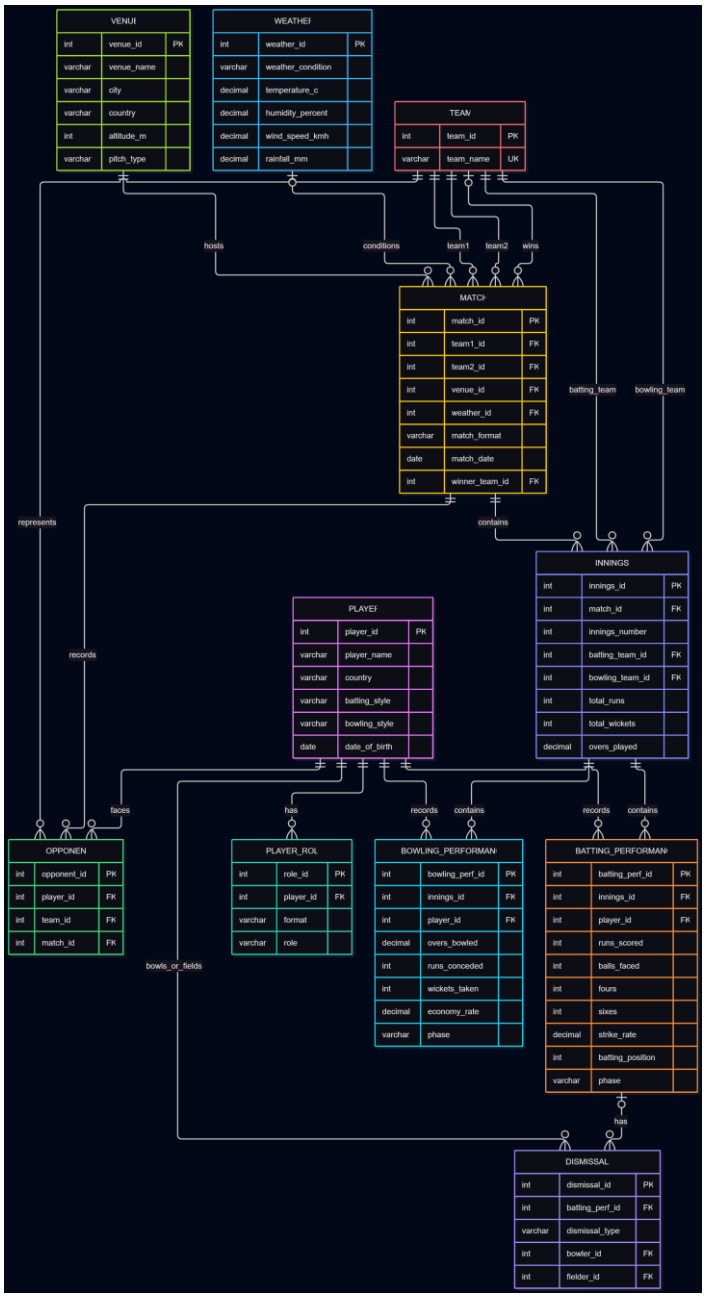


Figure 3.1. Entity–relationship model of the cricket selection database.

The ER diagram models links from teams, venue and weather to matches; matches to innings; innings to batting and bowling records; and players to roles and performances. Dismissal and opponent records reference the related player, performance, team and match data.

4. Relational Model

Relations and Entity Mapping

The relations map directly to the entities listed in the ER model. The project documentation identifies 12 MySQL tables and no separate associative relation outside the entity list.

Relational Schema

Relation	Key summary
player	player_id PK
team	team_id PK; team_name UNIQUE
match	match_id PK; team1_id, team2_id, venue_id, weather_id, winner_team_id FK
venue	venue_id PK
weather	weather_id PK
innings	innings_id PK; match_id, batting_team_id, bowling_team_id FK
player_role	role_id PK; player_id FK; UNIQUE(player_id, format)
batting_performance	batting_perf_id PK; innings_id, player_id FK
bowling_performance	bowling_perf_id PK; innings_id, player_id FK
dismissal	dismissal_id PK; batting_perf_id, bowler_id, fielder_id FK
opponent	opponent_id PK; player_id, team_id, match_id FK; UNIQUE(player_id, team_id, match_id)
users	user_id PK; email UNIQUE

Schema Diagram

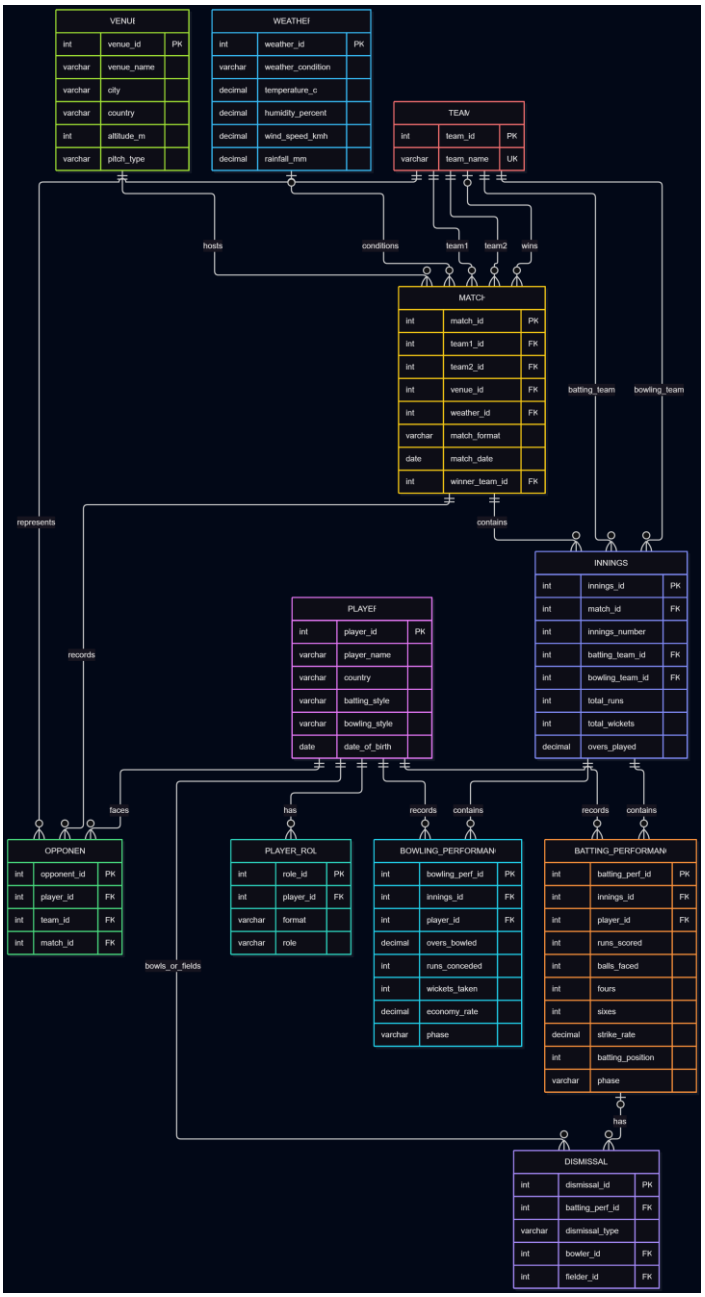


Figure 4.1. Relational schema and key relationships, represented in the supplied ER diagram.

The schema includes 12 tables, 19 documented foreign-key relationships and 2 views. The opponent relation records player–team–match associations. The supplied documentation does not designate a separate non-entity relation.

5. SQL DDL Statements

The MySQL database is named cricket_selection. DDL evidence in the source report shows database selection, table and constraint inspection, and schema creation. The schema defines primary keys, foreign keys, unique constraints, CHECK constraints, defaults and indexes. The full table definitions are evidenced by the preserved screenshots from the original report.

Database and Table Structure

```
USE cricket_selection;
SHOW TABLES;
DESC player;
```

The schema contains 12 tables: player, team, match, venue, weather, innings, player_role, batting_performance, bowling_performance, dismissal, opponent and users.

Keys, Constraints and ALTER Operations

The project schema includes primary keys on each table; foreign keys connecting match, innings and performance data; unique constraints on team names and user email addresses; and composite unique constraints on player_role and opponent. CHECK restrictions cover match formats, distinct teams, nonnegative performance values and other documented ranges. The SQL demonstration script shows CREATE TABLE, ALTER TABLE and DROP TABLE for a temporary sql_operations_demo table. The report describes example commands only where indicated; the temporary table is not represented as a permanent application table.



Figure 5.1. Preserved MySQL schema or DDL evidence from source report page 7.

```
mysql>
mysql> SELECT DATABASE();DESC player;
+--------------------+
| DATABASE()         |
+--------------------+
| cricket_selection  |
+--------------------+
1 row in set (0.00 sec)

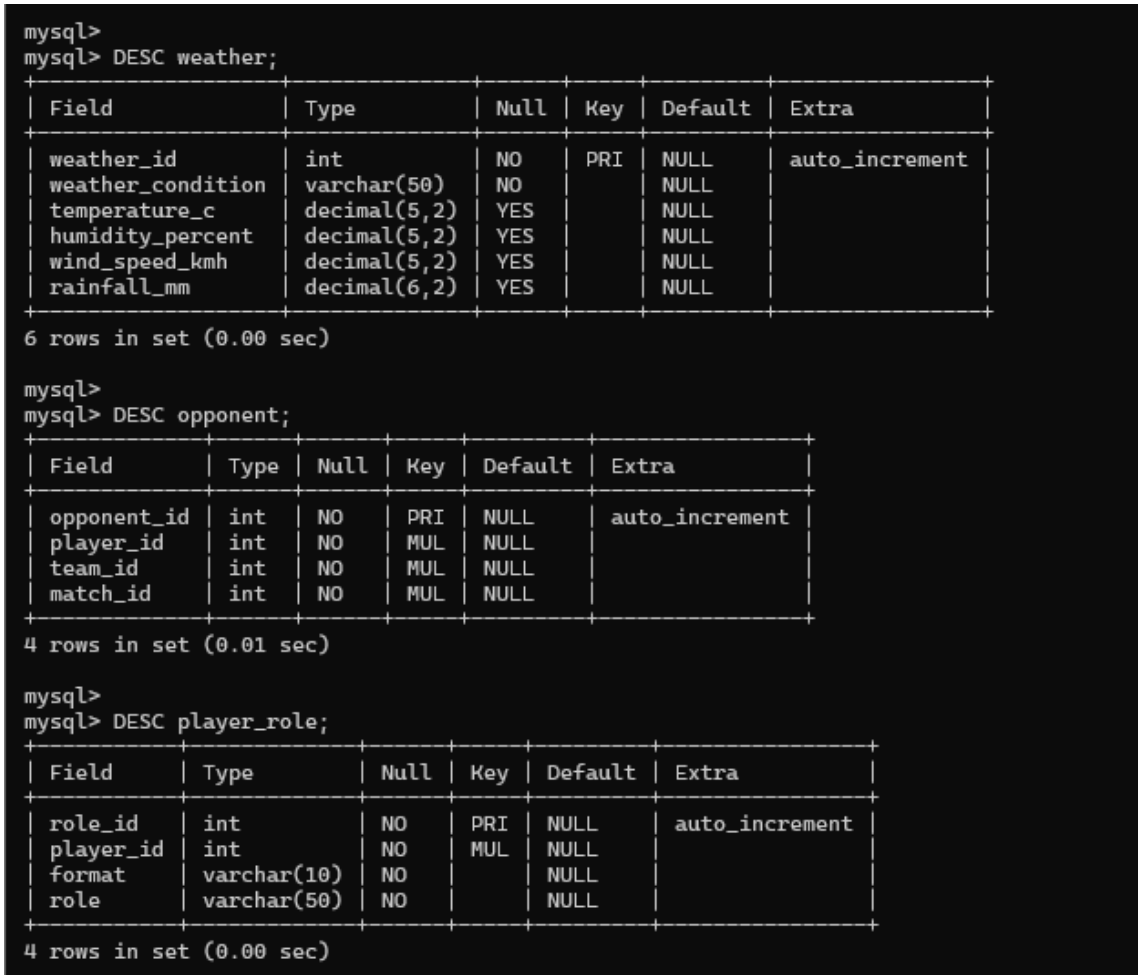
+---------------+--------------+------+-----+---------+----------------+
| Field         | Type         | Null | Key | Default | Extra          |
+---------------+--------------+------+-----+---------+----------------+
| player_id     | int          | NO   | PRI | NULL    | auto_increment |
| player_name   | varchar(100) | NO   |     | NULL    |                |
| country       | varchar(50)  | NO   |     | NULL    |                |
| batting_style | varchar(30)  | NO   |     | NULL    |                |
| bowling_style | varchar(50)  | YES  |     | NULL    |                |
| date_of_birth | date         | YES  |     | NULL    |                |
+---------------+--------------+------+-----+---------+----------------+
6 rows in set (0.01 sec)

mysql>
mysql> DESC team;
+-----------+-------------+------+-----+---------+----------------+
| Field     | Type        | Null | Key | Default | Extra          |
+-----------+-------------+------+-----+---------+----------------+
| team_id   | int         | NO   | PRI | NULL    | auto_increment |
| team_name | varchar(50) | NO   | UNI | NULL    |                |
+-----------+-------------+------+-----+---------+----------------+
2 rows in set (0.04 sec)

mysql>
mysql> DESC `match`;
+----------------+-------------+------+-----+---------+----------------+
| Field          | Type        | Null | Key | Default | Extra          |
+----------------+-------------+------+-----+---------+----------------+
| match_id       | int         | NO   | PRI | NULL    | auto_increment |
| team1_id       | int         | NO   | MUL | NULL    |                |
| team2_id       | int         | NO   | MUL | NULL    |                |
| venue_id       | int         | NO   | MUL | NULL    |                |
| weather_id     | int         | YES  | MUL | NULL    |                |
| match_format   | varchar(10) | NO   |     | NULL    |                |
| match_date     | date        | NO   |     | NULL    |                |
| winner_team_id | int         | YES  | MUL | NULL    |                |
+----------------+-------------+------+-----+---------+----------------+
8 rows in set (0.00 sec)

mysql>
mysql> DESC innings;
+-----------------+--------------+------+-----+---------+----------------+
| Field           | Type         | Null | Key | Default | Extra          |
+-----------------+--------------+------+-----+---------+----------------+
| innings_id      | int          | NO   | PRI | NULL    | auto_increment |
| match_id        | int          | NO   | MUL | NULL    |                |
| innings_number  | int          | NO   |     | NULL    |                |
| batting_team_id | int          | NO   | MUL | NULL    |                |
| bowling_team_id | int          | NO   | MUL | NULL    |                |
| total_runs      | int          | NO   |     | 0       |                |
| total_wickets   | int          | NO   |     | 0       |                |
| overs_played    | decimal(4,1) | YES  |     | NULL    |                |
+-----------------+--------------+------+-----+---------+----------------+
8 rows in set (0.00 sec)
```

Figure 5.2. Preserved MySQL schema or DDL evidence from source report page 8.



```
mysql>
mysql> DESC weather;
+--------------------+--------------+------+-----+---------+----------------+
| Field              | Type         | Null | Key | Default | Extra          |
+--------------------+--------------+------+-----+---------+----------------+
| weather_id         | int          | NO   | PRI | NULL    | auto_increment |
| weather_condition  | varchar(50)  | NO   |     | NULL    |                |
| temperature_c      | decimal(5,2) | YES  |     | NULL    |                |
| humidity_percent   | decimal(5,2) | YES  |     | NULL    |                |
| wind_speed_kmh     | decimal(5,2) | YES  |     | NULL    |                |
| rainfall_mm        | decimal(6,2) | YES  |     | NULL    |                |
+--------------------+--------------+------+-----+---------+----------------+
6 rows in set (0.00 sec)

mysql>
mysql> DESC opponent;
+-------------+------+------+-----+---------+----------------+
| Field       | Type | Null | Key | Default | Extra          |
+-------------+------+------+-----+---------+----------------+
| opponent_id | int  | NO   | PRI | NULL    | auto_increment |
| player_id   | int  | NO   | MUL | NULL    |                |
| team_id     | int  | NO   | MUL | NULL    |                |
| match_id    | int  | NO   | MUL | NULL    |                |
+-------------+------+------+-----+---------+----------------+
4 rows in set (0.01 sec)

mysql>
mysql> DESC player_role;
+-----------+-------------+------+-----+---------+----------------+
| Field     | Type        | Null | Key | Default | Extra          |
+-----------+-------------+------+-----+---------+----------------+
| role_id   | int         | NO   | PRI | NULL    | auto_increment |
| player_id | int         | NO   | MUL | NULL    |                |
| format    | varchar(10) | NO   |     | NULL    |                |
| role      | varchar(50) | NO   |     | NULL    |                |
+-----------+-------------+------+-----+---------+----------------+
4 rows in set (0.00 sec)
```

Figure 5.3. Preserved MySQL schema or DDL evidence from source report page 9.

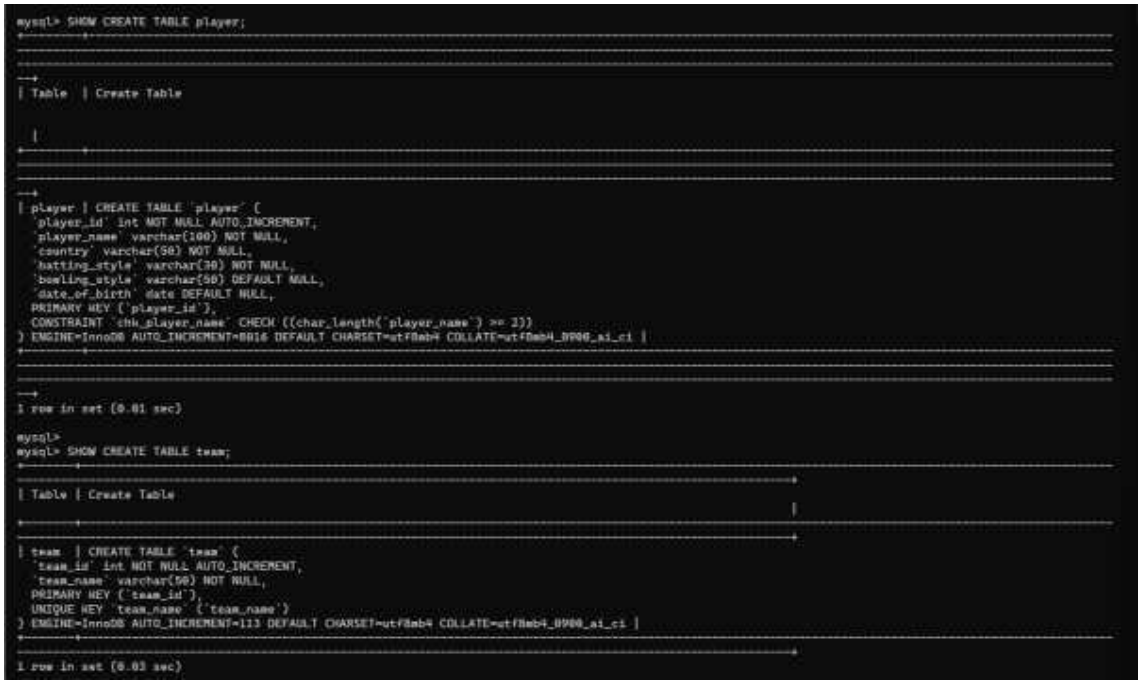
```
mysql>
mysql> DESC batting_performance;
+------------------+---------------+------+-----+---------+----------------+
| Field            | Type          | Null | Key | Default | Extra          |
+------------------+---------------+------+-----+---------+----------------+
| batting_perf_id  | int           | NO   | PRI | NULL    | auto_increment |
| innings_id       | int           | NO   | MUL | NULL    |                |
| player_id        | int           | NO   | MUL | NULL    |                |
| runs_scored      | int           | NO   |     | 0       |                |
| balls_faced      | int           | NO   |     | 0       |                |
| fours            | int           | NO   |     | 0       |                |
| sixes            | int           | NO   |     | 0       |                |
| strike_rate      | decimal(6,2)  | YES  |     | NULL    |                |
| batting_position | int           | YES  |     | NULL    |                |
| phase            | varchar(20)   | YES  |     | NULL    |                |
+------------------+---------------+------+-----+---------+----------------+
10 rows in set (0.01 sec)

mysql>
mysql> DESC bowling_performance;
+-----------------+---------------+------+-----+---------+----------------+
| Field           | Type          | Null | Key | Default | Extra          |
+-----------------+---------------+------+-----+---------+----------------+
| bowling_perf_id | int           | NO   | PRI | NULL    | auto_increment |
| innings_id      | int           | NO   | MUL | NULL    |                |
| player_id       | int           | NO   | MUL | NULL    |                |
| overs_bowled    | decimal(4,1)  | NO   |     | 0.0     |                |
| runs_conceded   | int           | NO   |     | 0       |                |
| wickets_taken   | int           | NO   |     | 0       |                |
| economy_rate    | decimal(5,2)  | YES  |     | NULL    |                |
| phase           | varchar(20)   | YES  |     | NULL    |                |
+-----------------+---------------+------+-----+---------+----------------+
8 rows in set (0.01 sec)

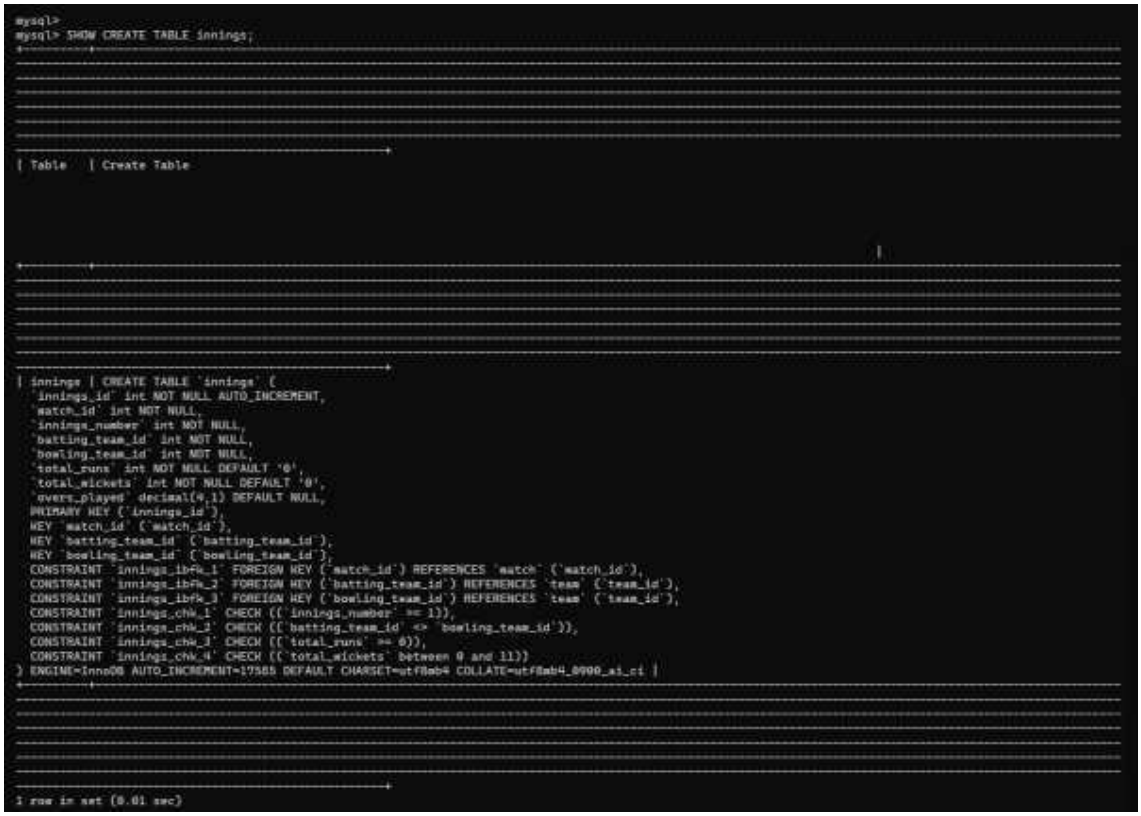
mysql>
mysql> DESC dismissal;
+-----------------+-------------+------+-----+---------+----------------+
| Field           | Type        | Null | Key | Default | Extra          |
+-----------------+-------------+------+-----+---------+----------------+
| dismissal_id    | int         | NO   | PRI | NULL    | auto_increment |
| batting_perf_id | int         | YES  | MUL | NULL    |                |
| dismissal_type  | varchar(30) | NO   |     | NULL    |                |
| bowler_id       | int         | YES  | MUL | NULL    |                |
| fielder_id      | int         | YES  | MUL | NULL    |                |
+-----------------+-------------+------+-----+---------+----------------+
5 rows in set (0.00 sec)

mysql>
mysql> DESC users;
+---------------+-----------------------+------+-----+-------------------+-------------------+
| Field         | Type                  | Null | Key | Default           | Extra             |
+---------------+-----------------------+------+-----+-------------------+-------------------+
| user_id       | int                   | NO   | PRI | NULL              | auto_increment    |
| name          | varchar(100)          | NO   |     | NULL              |                   |
| email         | varchar(150)          | NO   | UNI | NULL              |                   |
| password_hash | varchar(255)          | NO   |     | NULL              |                   |
| role          | enum('admin','analyst') | NO   |     | analyst           |                   |
| created_at    | timestamp             | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+---------------+-----------------------+------+-----+-------------------+-------------------+
6 rows in set (0.03 sec)
```

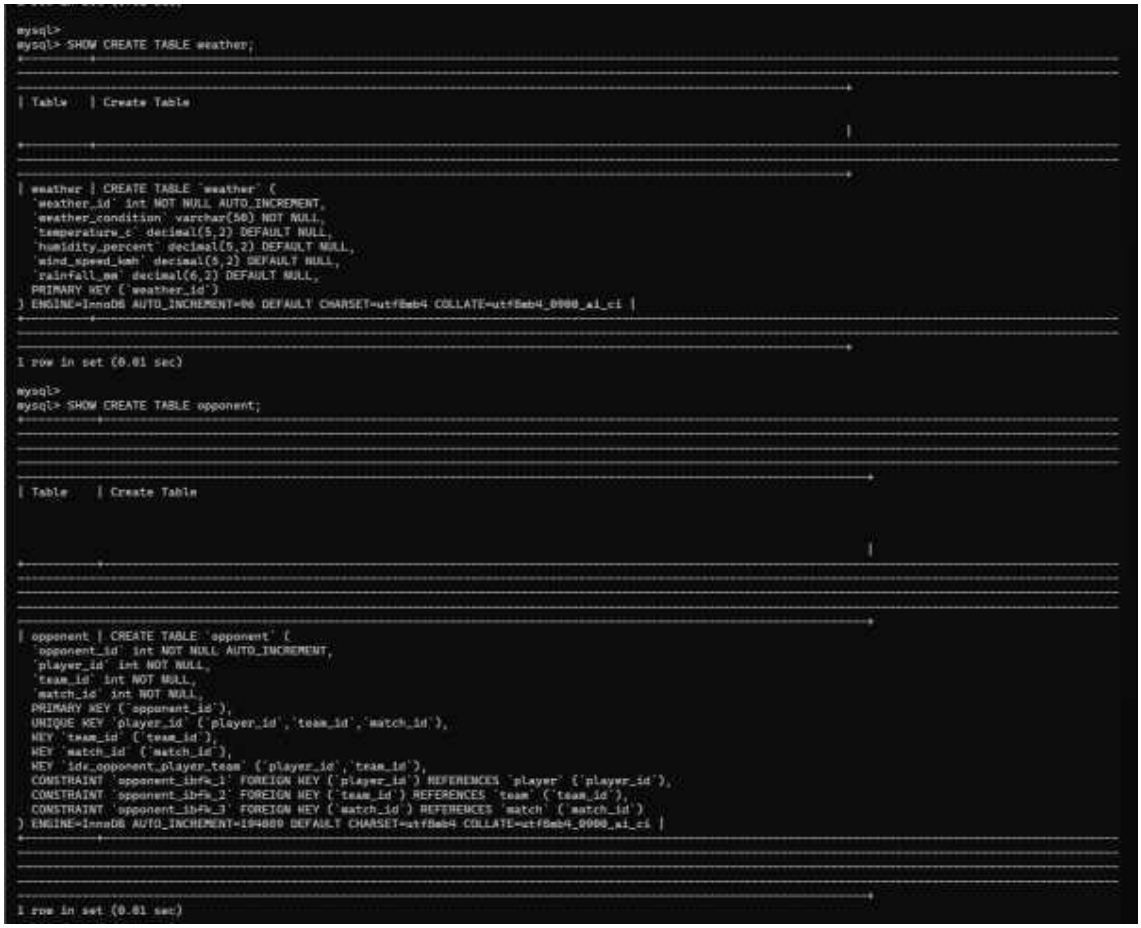
Figure 5.4. Preserved MySQL schema or DDL evidence from source report page 10.



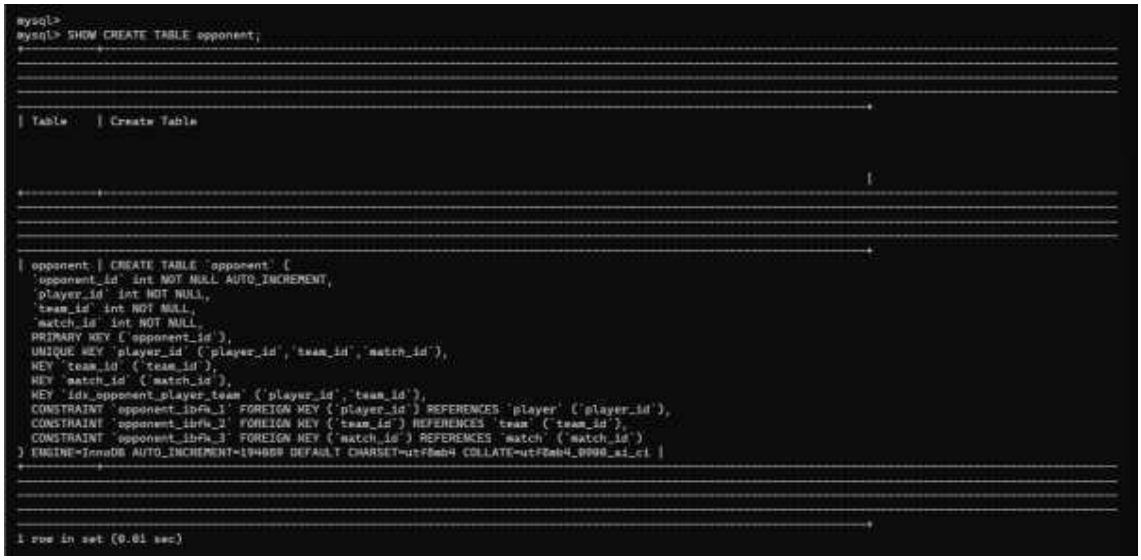
```
mysql> SHOW CREATE TABLE player;
+--------+-----------------------
```



```
mysql>
mysql> SHOW CREATE TABLE innings;
+---------+-----------------------
```



```
mysql>
mysql> SHOW CREATE TABLE weather;
+---------+-----------------------
```



```
mysql>
mysql> SHOW CREATE TABLE opponent;
+----------+-----------------------
```

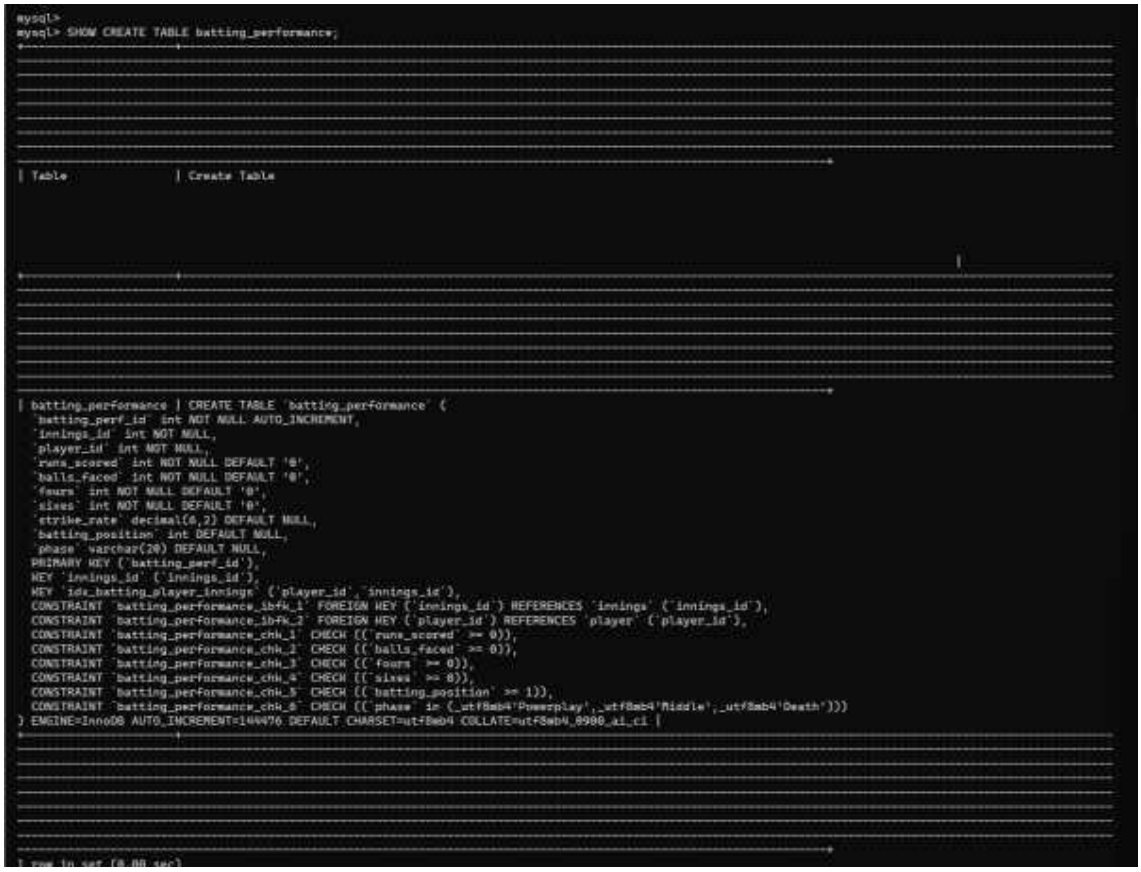
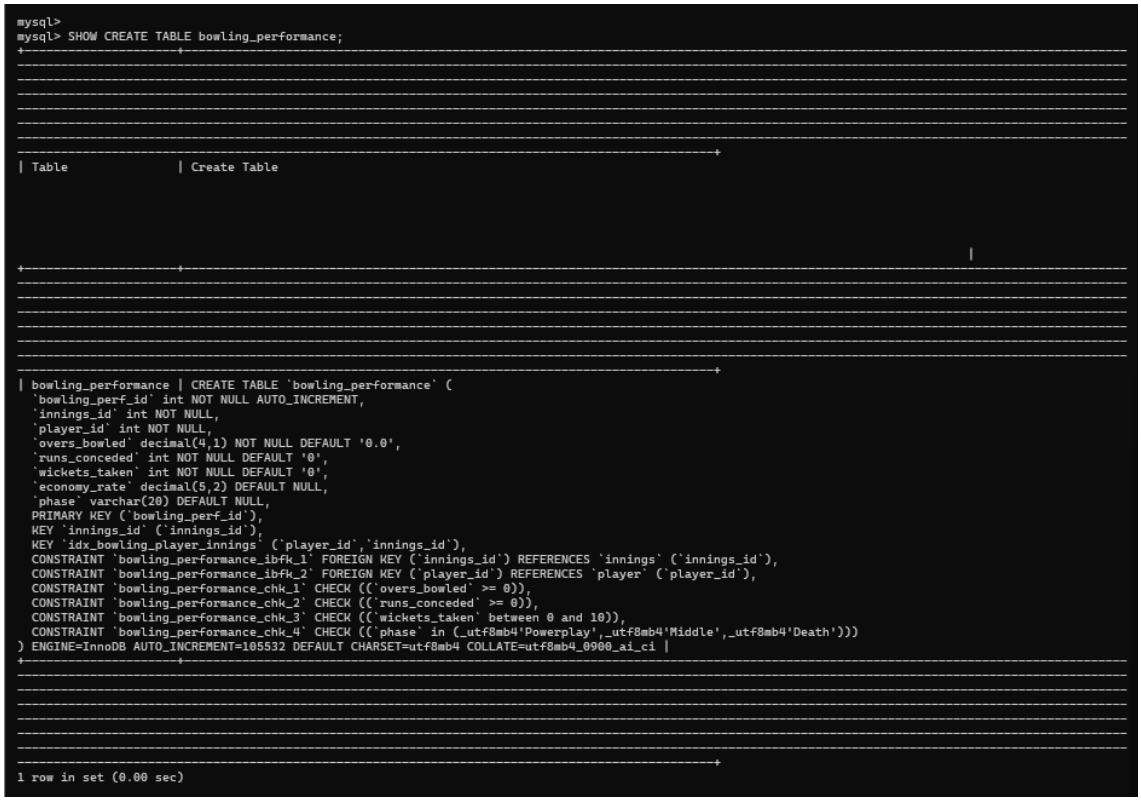
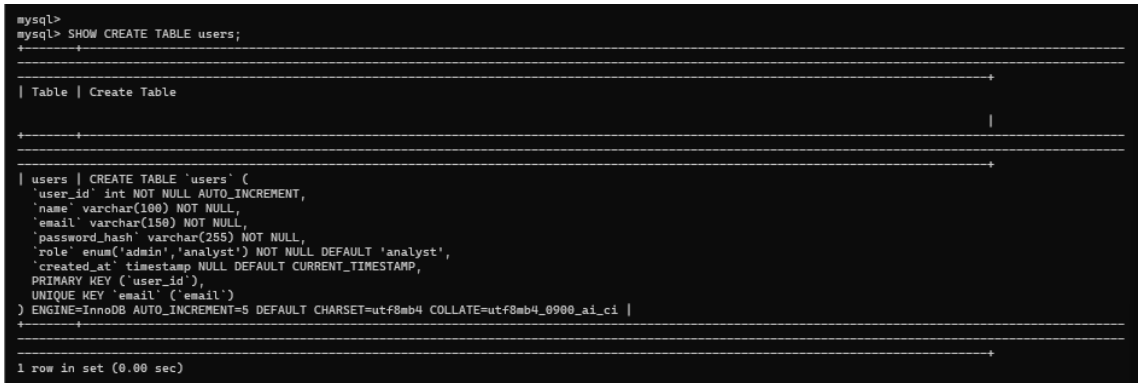


Figure 5.12. Preserved MySQL schema or DDL evidence from source report page 15.



```
mysql>
mysql> SHOW CREATE TABLE bowling_performance;
+----------------------+-----------------------
```



```
mysql>
mysql> SHOW CREATE TABLE users;
+-------+-----------------------
```

```
SELECT player_id, player_name FROM player WHERE player_id IN (
 SELECT player_id FROM batting_performance GROUP BY player_id
 HAVING SUM(runs_scored) > 5000);
```

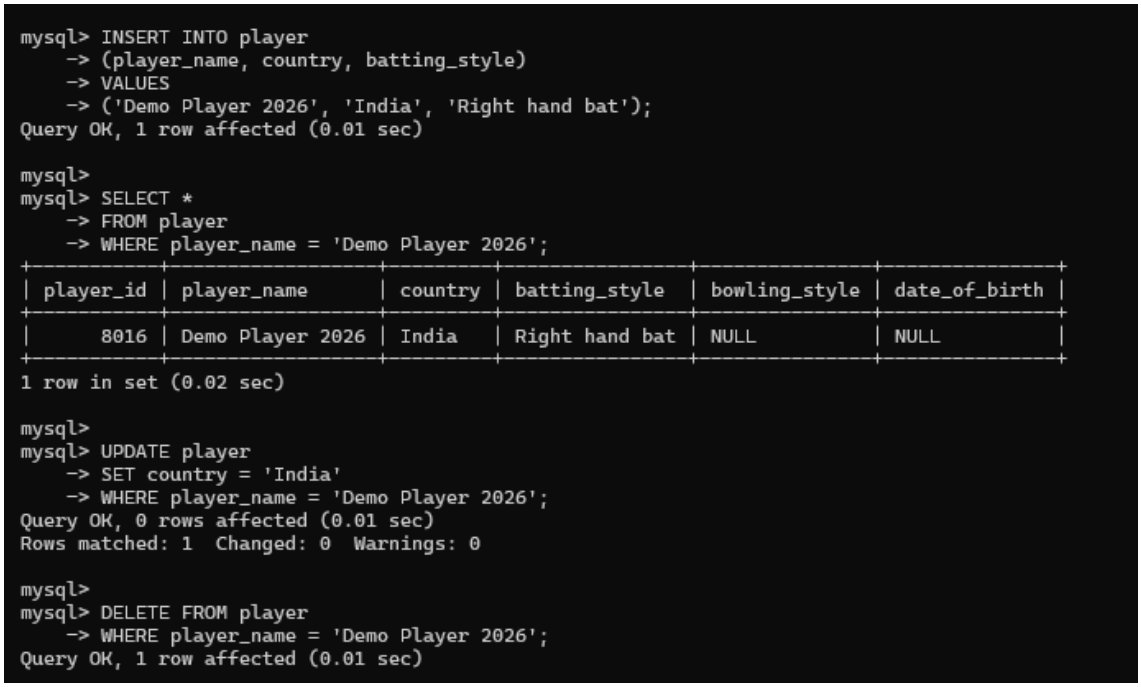
Views, Stored Procedure and Trigger

The supplied SQL files define vw_batting_performance and vw_player_details, the GetPlayerBattingStats procedure, and the trg_validate_batting_performance BEFORE INSERT trigger. The trigger rejects negative runs, balls faced, fours or sixes. The procedure accepts a player-name input and returns batting totals.

```
CALL GetPlayerBattingStats('Virat Kohli');
```

Indexes, EXPLAIN and Transactions

The database design notes document composite indexes idx_batting_player_innings, idx_bowling_player_innings, idx_match_venue_format and idx_opponent_player_team, in addition to primary-key, foreign-key and unique indexes. Index screenshots in the source report are retained below. The source report includes an indexes and EXPLAIN heading and a transactions/ACID discussion. The README also identifies CTEs and window-function ranking with RANK(). No user-defined SQL function is documented in the supplied scripts.



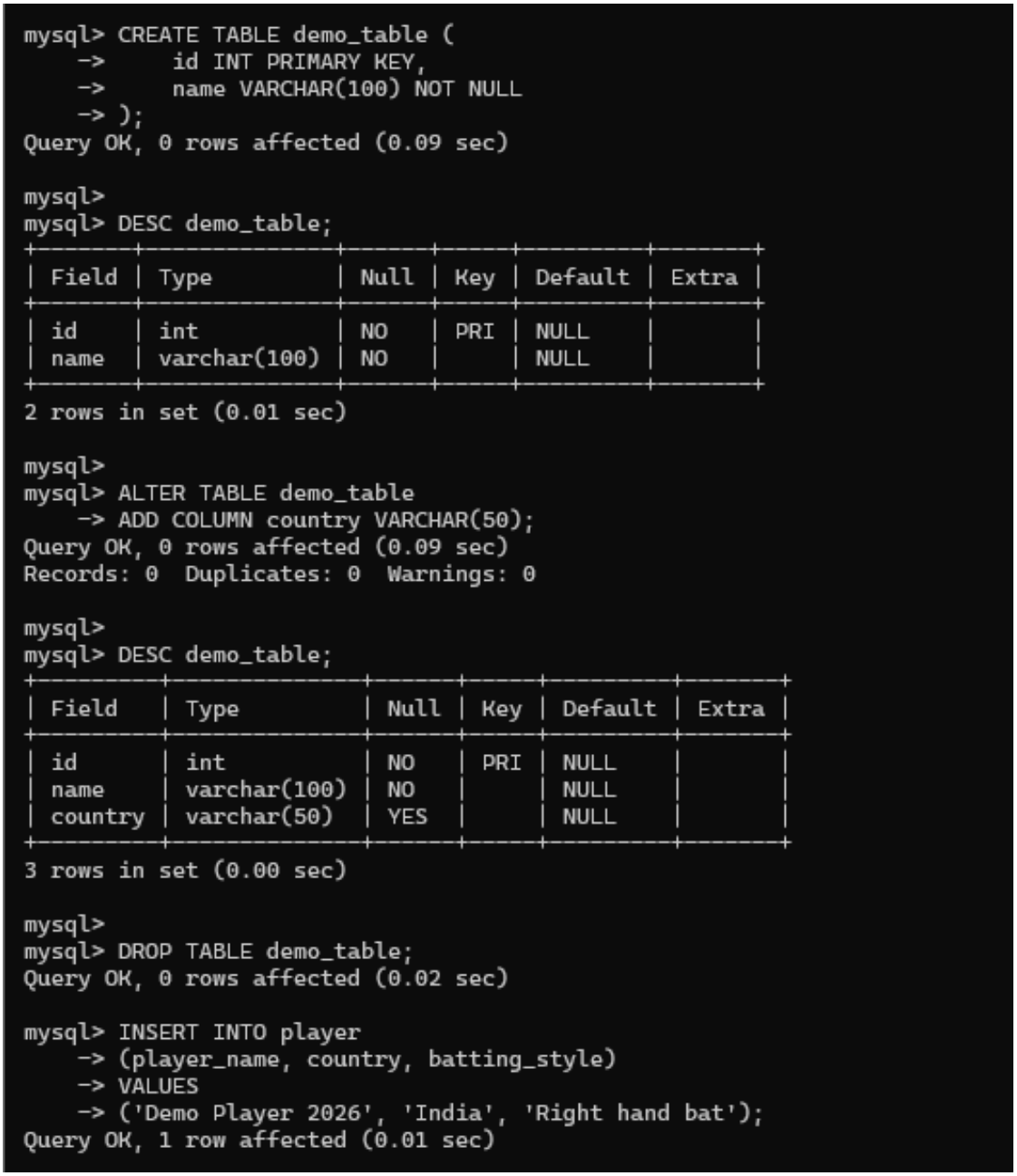
```
mysql> INSERT INTO player
    -> (player_name, country, batting_style)
    -> VALUES
    -> ('Demo Player 2026', 'India', 'Right hand bat');
Query OK, 1 row affected (0.01 sec)

mysql>
mysql> SELECT *
    -> FROM player
    -> WHERE player_name = 'Demo Player 2026';
+-----------+------------------+---------+----------------+---------------+---------------+
| player_id | player_name      | country | batting_style  | bowling_style | date_of_birth |
+-----------+------------------+---------+----------------+---------------+---------------+
|      8016 | Demo Player 2026 | India   | Right hand bat | NULL          | NULL          |
+-----------+------------------+---------+----------------+---------------+---------------+
1 row in set (0.02 sec)

mysql>
mysql> UPDATE player
    -> SET country = 'India'
    -> WHERE player_name = 'Demo Player 2026';
Query OK, 0 rows affected (0.01 sec)
Rows matched: 1  Changed: 0  Warnings: 0

mysql>
mysql> DELETE FROM player
    -> WHERE player_name = 'Demo Player 2026';
Query OK, 1 row affected (0.01 sec)
```

Figure 6.1. Preserved SQL operation, analytics or result evidence from source report page 17.



```
mysql> CREATE TABLE demo_table (
    ->     id INT PRIMARY KEY,
    ->     name VARCHAR(100) NOT NULL
    -> );
Query OK, 0 rows affected (0.09 sec)

mysql>
mysql> DESC demo_table;
+-------+--------------+------+-----+---------+-------+
| Field | Type         | Null | Key | Default | Extra |
+-------+--------------+------+-----+---------+-------+
| id    | int          | NO   | PRI | NULL    |       |
| name  | varchar(100) | NO   |     | NULL    |       |
+-------+--------------+------+-----+---------+-------+
2 rows in set (0.01 sec)

mysql>
mysql> ALTER TABLE demo_table
    -> ADD COLUMN country VARCHAR(50);
Query OK, 0 rows affected (0.09 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql>
mysql> DESC demo_table;
+---------+--------------+------+-----+---------+-------+
| Field   | Type         | Null | Key | Default | Extra |
+---------+--------------+------+-----+---------+-------+
| id      | int          | NO   | PRI | NULL    |       |
| name    | varchar(100) | NO   |     | NULL    |       |
| country | varchar(50)  | YES  |     | NULL    |       |
+---------+--------------+------+-----+---------+-------+
3 rows in set (0.00 sec)

mysql>
mysql> DROP TABLE demo_table;
Query OK, 0 rows affected (0.02 sec)

mysql> INSERT INTO player
    -> (player_name, country, batting_style)
    -> VALUES
    -> ('Demo Player 2026', 'India', 'Right hand bat');
Query OK, 1 row affected (0.01 sec)
```

Figure 6.2. Preserved SQL operation, analytics or result evidence from source report page 17.

```
mysql> SELECT *
    -> FROM player;
+-----------+--------------------------+-----------+----------------+-------------------------------------------+---------------+
| player_id | player_name              | country   | batting_style  | bowling_style                             | date_of_birth |
+-----------+--------------------------+-----------+----------------+-------------------------------------------+---------------+
|         1 | Virat Kohli              | India     | Right-handed   | NULL                                      | NULL          |
|         2 | M Klinger                | Australia | Right hand Bat |                                           | 1980-07-04    |
|         3 | TM Head                  | Australia | Left hand Bat  | Right arm Offbreak                        | 1993-12-29    |
|         4 | MC Henriques             | Australia | Right hand Bat | Right arm Fast medium                     | 1987-02-01    |
|         5 | AJ Turner                | Australia | Right hand Bat | Right arm Offbreak                        | 1993-01-25    |
|         6 | JP Faulkner              | Australia | Right hand Bat | Left arm Fast medium                      | 1990-04-29    |
|         7 | TD Paine                 | Australia | Right hand Bat | Right arm Medium                          | 1984-12-08    |
|         8 | PJ Cummins               | Australia | Right hand Bat | Right arm Fast                            | 1993-05-08    |
|         9 | A Zampa                  | Australia | Right hand Bat | Legbreak Googly                           | 1992-03-31    |
|        10 | B Stanlake               | Australia | Left hand Bat  | Right arm Fast                            | 1994-11-04    |
|        11 | AJ Tye                   | Australia | Right hand Bat | Right arm Medium fast                     | 1986-12-12    |
|        12 | N Dickwella              | Sri Lanka | Left hand Bat  |                                           | 1993-06-23    |
|        13 | WU Tharanga              | Sri Lanka | Left hand Bat  |                                           | 1985-02-02    |
|        14 | EMDY Munaweera           | Sri Lanka | Right hand Bat | Right arm Offbreak                        | 1989-04-24    |
|        15 | DAS Gunaratne            | Sri Lanka | Right hand Bat | Right arm Medium fast                     | 1986-01-08    |
|        16 | TAM Siriwardana          | Sri Lanka | Left hand Bat  | Slow Left arm Orthodox                    | 1985-12-04    |
|        17 | CK Kapugedera            | Sri Lanka | Right hand Bat | Right arm Medium                          | 1987-02-24    |
|        18 | S Prasanna               | Sri Lanka | Right hand Bat | Legbreak                                  | 1985-06-27    |
|        19 | JRMVB Sanjaya            | Sri Lanka | Right hand Bat | Right arm Fast medium                     | 1992-02-10    |
|        20 | SL Malinga               | Sri Lanka | Right hand Bat | Right arm Fast                            | 1983-08-28    |
|        21 | KMDN Kulasekara          | Sri Lanka | Right hand Bat | Right arm Fast medium                     | 1982-07-22    |
|        22 | PADLR Sandakan           | Sri Lanka | Right hand Bat | Left arm Wrist spin                       | 1991-06-10    |
|        23 | BR Dunk                  | Australia | Left hand Bat  | Right arm Offbreak                        | 1987-03-11    |
|        24 | JA Richardson            | Australia | Right hand Bat | Right arm Fast                            | 1996-09-20    |
|        25 | BKG Mendis               | Sri Lanka | Right hand Bat | Legbreak                                  | 1995-02-02    |
|        26 | MD Shanaka               | Sri Lanka | Right hand Bat | Right arm Medium                          | 1991-09-09    |
|        27 | Nizakat Khan             | Hong Kong | Right hand Bat | Legbreak                                  | 1992-07-08    |
|        28 | Aizaz Khan               | Hong Kong | Right hand Bat | Right arm Medium fast                     | 1993-03-21    |
|        29 | Babar Hayat              | Hong Kong | Right hand Bat | Right arm Medium                          | 1992-01-05    |
|        30 | Anshuman Rath            | Hong Kong | Left hand Bat  | Slow Left arm Orthodox                    | 1997-11-05    |
|        31 | Ehsan Khan               | Hong Kong | Right hand Bat | Right arm Offbreak                        | 1984-12-27    |
|        32 | Tanwir Afzal             | Hong Kong | Right hand Bat | Right arm Fast medium                     | 1988-06-12    |
|        33 | Shahid Wasif             | Hong Kong | Right hand Bat | Right arm Offbreak                        | 1997-05-15    |
|        34 | Tanveer Ahmed            | Hong Kong | Right hand Bat | Right arm Medium fast                     | 1997-09-18    |
|        35 | Ehsan Nawaz              | Hong Kong | Right hand Bat | Right arm Fast medium                     | 1995-03-21    |
|        36 | Nadeem Ahmed             | Hong Kong | Right hand Bat | Slow Left arm Orthodox                    | 1987-09-28    |
|        37 | C Carter                 | Hong Kong | Right hand Bat |                                           | 1997-09-09    |
|        38 | WTS Porterfield          | Ireland   | Left hand Bat  | Right arm Offbreak                        | 1984-09-06    |
|        39 | DA Rankin                | Ireland   | Right hand Bat | Right arm Offbreak                        | 1987-08-02    |
|        40 | SP Terry                 | Ireland   | Right hand Bat | Right arm Offbreak                        | 1991-08-01    |
|        41 | JNK Shannon              | Ireland   | Right hand Bat | Right arm Offbreak                        | 1990-02-12    |
|        42 | KJ O'Brien               | Ireland   | Right hand Bat | Right arm Medium fast                     | 1984-03-04    |
|        43 | GJ Thompson              | Ireland   | Right hand Bat | Legbreak                                  | 1987-09-17    |
|        44 | AR McBrine               | Ireland   | Left hand Bat  | Right arm Offbreak                        | 1993-04-30    |
|        45 | L Tucker                 | Ireland   | Right hand Bat |                                           | 1996-09-10    |
|        46 | J Mulder                 | Ireland   | Right hand Bat | Legbreak                                  | 1995-08-11    |
|        47 | J Little                 | Ireland   | Right hand Bat | Left arm Fast medium                      | 1999-11-01    |
|        48 | CA Young                 | Ireland   | Right hand Bat | Right arm Fast medium                     | 1990-04-04    |
|        49 | KL Rahul                 | India     | Right hand Bat |                                           | 1992-04-18    |
|        50 | Mandeep Singh            | India     | Right hand Bat | Right arm Medium                          | 1991-12-18    |
```

Figure 6.3. Preserved SQL operation, analytics or result evidence from source report page 18.

```
mysql>
mysql> SELECT player_id, player_name, country
    -> FROM player;
+-----------+-------------------------------+---------------------------+
| player_id | player_name                   | country                   |
+-----------+-------------------------------+---------------------------+
|         1 | Virat Kohli                   | India                     |
|         2 | M Klinger                     | Australia                 |
|         3 | TM Head                       | Australia                 |
|         4 | MC Henriques                  | Australia                 |
|         5 | AJ Turner                     | Australia                 |
|         6 | JP Faulkner                   | Australia                 |
|         7 | TD Paine                      | Australia                 |
|         8 | PJ Cummins                    | Australia                 |
|         9 | A Zampa                       | Australia                 |
|        10 | B Stanlake                    | Australia                 |
|        11 | AJ Tye                        | Australia                 |
|        12 | N Dickwella                   | Sri Lanka                 |
|        13 | WU Tharanga                   | Sri Lanka                 |
|        14 | EMDY Munaweera                | Sri Lanka                 |
|        15 | DAS Gunaratne                 | Sri Lanka                 |
|        16 | TAM Siriwardana               | Sri Lanka                 |
|        17 | CK Kapugedera                 | Sri Lanka                 |
|        18 | S Prasanna                    | Sri Lanka                 |
|        19 | JRMVB Sanjaya                 | Sri Lanka                 |
|        20 | SL Malinga                    | Sri Lanka                 |
|        21 | KMDN Kulasekara               | Sri Lanka                 |
|        22 | PADLR Sandakan                | Sri Lanka                 |
|        23 | BR Dunk                       | Australia                 |
|        24 | JA Richardson                 | Australia                 |
|        25 | BKG Mendis                    | Sri Lanka                 |
|        26 | MD Shanaka                    | Sri Lanka                 |
|        27 | Nizakat Khan                  | Hong Kong                 |
|        28 | Aizaz Khan                    | Hong Kong                 |
|        29 | Babar Hayat                   | Hong Kong                 |
|        30 | Anshuman Rath                 | Hong Kong                 |
|        31 | Ehsan Khan                    | Hong Kong                 |
|        32 | Tanwir Afzal                  | Hong Kong                 |
|        33 | Shahid Wasif                  | Hong Kong                 |
|        34 | Tanveer Ahmed                 | Hong Kong                 |
|        35 | Ehsan Nawaz                   | Hong Kong                 |
|        36 | Nadeem Ahmed                  | Hong Kong                 |
|        37 | C Carter                      | Hong Kong                 |
|        38 | WTS Porterfield               | Ireland                   |
|        39 | DA Rankin                     | Ireland                   |
|        40 | SP Terry                      | Ireland                   |
|        41 | JNK Shannon                   | Ireland                   |
|        42 | KJ O'Brien                    | Ireland                   |
|        43 | GJ Thompson                   | Ireland                   |
|        44 | AR McBrine                    | Ireland                   |
|        45 | L Tucker                      | Ireland                   |
|        46 | J Mulder                      | Ireland                   |
|        47 | J Little                      | Ireland                   |
|        48 | CA Young                      | Ireland                   |
|        49 | KL Rahul                      | India                     |
|        50 | Mandeep Singh                 | India                     |
```

Figure 6.4. Preserved SQL operation, analytics or result evidence from source report page 19.

```
014 rows in set (0.mysql>
mysql> SELECT player_name, country
    -> FROM player
    -> WHERE country = 'India';
+----------------------+---------+
| player_name          | country |
+----------------------+---------+
| Virat Kohli          | India   |
| KL Rahul             | India   |
| Mandeep Singh        | India   |
| AT Rayudu            | India   |
| MK Pandey            | India   |
| KM Jadhav            | India   |
| MS Dhoni             | India   |
| AR Patel             | India   |
| R Dhawan             | India   |
| JJ Bumrah            | India   |
| JD Unadkat           | India   |
| YS Chahal            | India   |
| DS Kulkarni          | India   |
| BB Sran              | India   |
| V Kohli              | India   |
| SK Raina             | India   |
| Yuvraj Singh         | India   |
| HH Pandya            | India   |
| Parvez Rasool        | India   |
| A Nehra              | India   |
| RR Pant              | India   |
| RG Sharma            | India   |
| AM Rahane            | India   |
| STR Binny            | India   |
| R Ashwin             | India   |
| RA Jadeja            | India   |
| Mohammed Shami       | India   |
| B Kumar              | India   |
| M Raj                | India   |
| S Mandhana           | India   |
| S Meghana            | India   |
| V Krishnamurthy      | India   |
| H Kaur               | India   |
| J Goswami            | India   |
| AA Patil             | India   |
| S Pandey             | India   |
| Preeti Bose          | India   |
| S Verma              | India   |
| E Bisht              | India   |
| A Poulose            | India   |
```

Figure 6.5. Preserved SQL operation, analytics or result evidence from source report page 20.



```
mysql>
mysql> uery aborted
mysql> SELECT DISTINCT country
    -> FROM player;
+--------------------------+
| country                  |
+--------------------------+
| India                    |
| Australia                |
| Sri Lanka                |
| Hong Kong                |
| Ireland                  |
| Zimbabwe                 |
| Bangladesh               |
| New Zealand              |
| South Africa             |
| England                  |
| West Indies              |
| United States of America |
| Pakistan                 |
| Scotland                 |
| Netherlands              |
| Oman                     |
| United Arab Emirates     |
| Afghanistan              |
| Papua New Guinea         |
| Nepal                    |
| Thailand                 |
| Uganda                   |
| Malaysia                 |
| Botswana                 |
| Lesotho                  |
| Malawi                   |
| Namibia                  |
| Sierra Leone             |
| Mozambique               |
| China                    |
| Kuwait                   |
| Philippines              |
| Vanuatu                  |
| Germany                  |
| Italy                    |
| Nigeria                  |
| Tanzania                 |
| Rwanda                   |
| Kenya                    |
| Japan                    |
| Indonesia                |
| Fiji                     |
| Samoa                    |
| Canada                   |
| Ghana                    |
| Guernsey                 |
| Denmark                  |
| Norway                   |
| Jersey                   |
| Maldives                 |
| Mali                     |
| Singapore                |
```

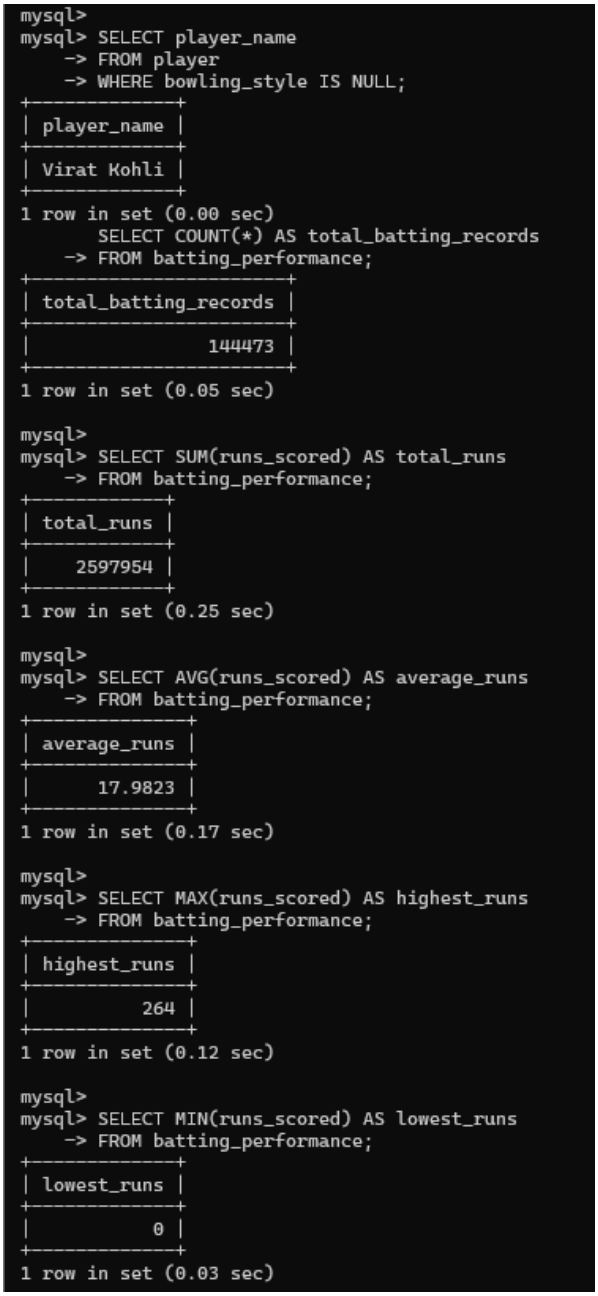
Figure 6.6. Preserved SQL operation, analytics or result evidence from source report page 21.

```
mysql>
mysql> SELECT player_name
    -> FROM player
    -> ORDER BY player_name ASC;
+-------------------------------+
| player_name                   |
+-------------------------------+
| A Adekunle                    |
| A Ahmadhel                    |
| A Aikenhead                   |
| A Aitken                      |
| A Aitken-Drummond             |
| A Akhigbe                     |
| A Akurut                      |
| A Alexander                   |
| A Alfonso                     |
| A Amado                       |
| A Amburose                    |
| A Amuzu                       |
| A Andrews                     |
| A Andriani                    |
| A Argyropoulou                |
| A Ashok                       |
| A Ashokan                     |
| A Athanaze                    |
| A Augastin                    |
| A Awasthi                     |
| A Bagai                       |
| A Balbirnie                   |
| A Beggs                       |
| A Beresford-Peirse            |
| A Bester                      |
| A Bhadauria                   |
| A Bhagwat                     |
| A Bhargava                    |
| A Bhattarai                   |
| A Bidari                      |
| A Bierwisch                   |
| A Bishnoi                     |
| A Bohara                      |
| A Bosch                       |
| A Bose                        |
| A Brindle                     |
| A Brown-Carrera               |
| A Bulow                       |
| A Bushell                     |
| A Busing-Volschenk            |
| A Canning                     |
| A Capsey                      |
| A Cespedes                    |
| A Chandel                     |
| A Charles                     |
| A Chaubey                     |
| A Chaudhary                   |
| A Chenthamara                 |
| A Cheung                      |
| A Chilia                      |
| A Chopra                      |
| A Codrington                  |
| A Conteh                      |
| A Cossa                       |
| A Curavic                     |
| A Dalzell                     |
| A Dananjaya                   |
| A Das                         |
| A Dastgir                     |
| A Davidson Soler              |
| A Davidson-Soler              |
| A Daxini                      |
| A Dcosta                      |
| A Denuwan                     |
| A Dercksen                    |
| A Desai                       |
| A Dhabe                       |
| A Dhar                        |
```

Figure 6.7. Preserved SQL operation, analytics or result evidence from source report page 22.

```
mysql>
mysql> SELECT *
    -> FROM batting_performance
    -> LIMIT 10;
+-----------------+-------------+-----------+-------------+-------------+-------+-------+-------------+------------------+------------+
| batting_perf_id | innings_id | player_id | runs_scored | balls_faced | fours | sixes | strike_rate | batting_position | phase      |
+-----------------+-------------+-----------+-------------+-------------+-------+-------+-------------+------------------+------------+
|               1 |           1 |         1 |          43 |          34 |     2 |     2 |      126.47 |                1 | Middle     |
|               2 |           1 |         2 |          38 |          32 |     4 |     0 |      118.75 |                2 | Powerplay  |
|               3 |           1 |         3 |          31 |          25 |     0 |     1 |      124.00 |                3 | Middle     |
|               4 |           1 |         4 |          17 |          10 |     0 |     1 |      170.00 |                4 | Middle     |
|               5 |           1 |         5 |          18 |          12 |     1 |     0 |      150.00 |                5 | Middle     |
|               6 |           1 |         6 |          14 |           7 |     2 |     0 |      200.00 |                6 | Death      |
|               7 |           2 |        12 |          30 |          25 |     4 |     1 |      120.00 |                1 | Powerplay  |
|               8 |           2 |        13 |           0 |           2 |     0 |     0 |        0.00 |                2 | Powerplay  |
|               9 |           2 |        14 |          44 |          29 |     6 |     0 |      151.72 |                3 | Powerplay  |
|              10 |           2 |        15 |          52 |          37 |     7 |     0 |      140.54 |                4 | Middle     |
+-----------------+-------------+-----------+-------------+-------------+-------+-------+-------------+------------------+------------+
10 rows in set (0.00 sec)
```

Figure 6.8. Preserved SQL operation, analytics or result evidence from source report page 22.



```
mysql>
mysql> SELECT player_name
    -> FROM player
    -> WHERE bowling_style IS NULL;
+-------------+
| player_name |
+-------------+
| Virat Kohli |
+-------------+
1 row in set (0.00 sec)
       SELECT COUNT(*) AS total_batting_records
    -> FROM batting_performance;
+-----------------------+
| total_batting_records |
+-----------------------+
|                144473 |
+-----------------------+
1 row in set (0.05 sec)

mysql>
mysql> SELECT SUM(runs_scored) AS total_runs
    -> FROM batting_performance;
+------------+
| total_runs |
+------------+
|    2597954 |
+------------+
1 row in set (0.25 sec)

mysql>
mysql> SELECT AVG(runs_scored) AS average_runs
    -> FROM batting_performance;
+--------------+
| average_runs |
+--------------+
|      17.9823 |
+--------------+
1 row in set (0.17 sec)

mysql>
mysql> SELECT MAX(runs_scored) AS highest_runs
    -> FROM batting_performance;
+--------------+
| highest_runs |
+--------------+
|          264 |
+--------------+
1 row in set (0.12 sec)

mysql>
mysql> SELECT MIN(runs_scored) AS lowest_runs
    -> FROM batting_performance;
+-------------+
| lowest_runs |
+-------------+
|           0 |
+-------------+
1 row in set (0.03 sec)
```

Figure 6.9. Preserved SQL operation, analytics or result evidence from source report page 23.

```
mysql> SELECT
    ->     p.player_id,
    ->     p.player_name,
    ->     SUM(bp.runs_scored) AS total_runs
    -> FROM player p
    -> JOIN batting_performance bp
    ->     ON p.player_id = bp.player_id
    -> GROUP BY p.player_id, p.player_name
    -> ORDER BY total_runs DESC;
+-----------+-----------------------------+------------+
| player_id | player_name                 | total_runs |
+-----------+-----------------------------+------------+
|       157 | V Kohli                     |      18788 |
|       165 | RG Sharma                   |      15574 |
|      6822 | KC Sangakkara               |      13022 |
|        54 | MS Dhoni                    |      11858 |
|       110 | AB de Villiers              |      11026 |
|      6821 | TM Dilshan                  |      10959 |
|       219 | Babar Azam                  |      10873 |
|       257 | MJ Guptill                  |      10642 |
|       258 | LRPL Taylor                 |      10100 |
|       108 | Q de Kock                   |      10000 |
|       259 | DA Warner                   |       9877 |
|       122 | JC Buttler                  |       9710 |
|        90 | KS Williamson               |       9682 |
|      6820 | DPMD Jayawardene            |       9531 |
|       121 | EJG Morgan                  |       9396 |
|        78 | Tamim Iqbal                 |       9272 |
|       107 | HM Amla                     |       9106 |
|       246 | S Mandhana                  |       8968 |
|        81 | Shakib Al Hasan             |       8933 |
|       335 | Mohammad Hafeez             |       8788 |
|       372 | S Dhawan                    |       8492 |
|       151 | JE Root                     |       8459 |
|       328 | Mushfiqur Rahim             |       8403 |
|       220 | Shoaib Malik                |       8373 |
|       140 | CH Gayle                    |       8365 |
|         1 | Virat Kohli                 |       8308 |
|       468 | L Wolvaardt                 |       8217 |
|       159 | Yuvraj Singh                |       8068 |
|       382 | SD Hope                     |       7531 |
|      6756 | MJ Clarke                   |       7476 |
|      6758 | RT Ponting                  |       7380 |
|        83 | Mahmudullah                 |       7354 |
|       131 | DA Miller                   |       7351 |
|      6776 | BB McCullum                 |       7302 |
|       158 | SK Raina                    |       7134 |
|        25 | BKG Mendis                  |       7001 |
|       375 | AD Mathews                  |       6989 |
|       109 | F du Plessis                |       6932 |
|       179 | BL Mooney                   |       6894 |
|        13 | WU Tharanga                 |       6889 |
|       208 | AC Jayangani                |       6863 |
|       424 | SR Taylor                   |       6850 |
|      6846 | SR Watson                   |       6822 |
|       111 | JP Duminy                   |       6808 |
|      1173 | SPD Smith                   |       6758 |
|       307 | PR Stirling                 |       6708 |
|       249 | H Kaur                      |       6668 |
|       189 | SW Bates                    |       6596 |
```

Figure 6.10. Preserved SQL operation, analytics or result evidence from source report page 24.

```
mysql> SELECT
    ->     p.player_id,
    ->     p.player_name,
    ->     SUM(bp.runs_scored) AS total_runs
    -> FROM player p
    -> JOIN batting_performance bp
    ->     ON p.player_id = bp.player_id
    -> GROUP BY p.player_id, p.player_name
    -> HAVING SUM(bp.runs_scored) > 1000;
+-----------+----------------------+------------+
| player_id | player_name          | total_runs |
+-----------+----------------------+------------+
|         1 | Virat Kohli          |       8308 |
|         3 | TM Head              |       4283 |
|         6 | JP Faulkner          |       1184 |
|        12 | N Dickwella          |       2084 |
|        13 | WU Tharanga          |       6889 |
|        17 | CK Kapugedera        |       2180 |
|        25 | BKG Mendis           |       7001 |
|        21 | KMDN Kulasekara      |       1519 |
|        26 | MD Shanaka           |       3133 |
|        27 | Nizakat Khan         |       2896 |
|        28 | Aizaz Khan           |       1218 |
|        29 | Babar Hayat          |       2954 |
|        30 | Anshuman Rath        |       2700 |
|        38 | WTS Porterfield      |       3390 |
|        42 | KJ O'Brien           |       3561 |
|        45 | L Tucker             |       2420 |
|        60 | CJ Chibhabha         |       2499 |
|        61 | H Masakadza          |       5185 |
|        63 | Sikandar Raza        |       6392 |
|        64 | MN Waller            |       1546 |
|        66 | E Chigumbura         |       4035 |
|        49 | KL Rahul             |       5432 |
|        51 | AT Rayudu            |       1679 |
|        52 | MK Pandey            |       1267 |
|        53 | KM Jadhav            |       1440 |
|        54 | MS Dhoni             |      11858 |
|        55 | AR Patel             |       1656 |
|        75 | V Sibanda            |       2623 |
|        78 | Tamim Iqbal          |       9272 |
|        79 | Imrul Kayes          |       2361 |
|        80 | Sabbir Rahman        |       2198 |
|        81 | Shakib Al Hasan      |       8933 |
|        82 | Soumya Sarkar        |       3803 |
|        83 | Mahmudullah          |       7354 |
|        85 | Mashrafe Mortaza     |       1595 |
|        90 | KS Williamson        |       9682 |
|        91 | C Munro              |       2978 |
|        92 | CJ Anderson          |       1799 |
|        94 | C de Grandhomme      |       1247 |
|        95 | L Ronchi             |       1756 |
|       100 | JDS Neesham          |       2613 |
|        96 | MJ Santner           |       2728 |
|       107 | HM Amla              |       9106 |
|       108 | Q de Kock            |      10000 |
```

Figure 6.11. Preserved SQL operation, analytics or result evidence from source report page 25.

```
mysql> SELECT p.player_name, bp.runs_scored
    -> FROM player p
    -> INNER JOIN batting_performance bp
    -> ON p.player_id = bp.player_id;
+--------------------------------+-------------+
| player_name                    | runs_scored |
+--------------------------------+-------------+
| Virat Kohli                    |          43 |
| Virat Kohli                    |          12 |
| Virat Kohli                    |          53 |
| Virat Kohli                    |          20 |
| Virat Kohli                    |          36 |
| Virat Kohli                    |          18 |
| Virat Kohli                    |          42 |
| Virat Kohli                    |           8 |
| Virat Kohli                    |          84 |
| Virat Kohli                    |          68 |
| Virat Kohli                    |         172 |
| Virat Kohli                    |          16 |
| Virat Kohli                    |           3 |
| Virat Kohli                    |          47 |
| Virat Kohli                    |           7 |
| Virat Kohli                    |          27 |
| Virat Kohli                    |           0 |
| Virat Kohli                    |          28 |
| Virat Kohli                    |           0 |
| Virat Kohli                    |           3 |
| Virat Kohli                    |           1 |
| Virat Kohli                    |           1 |
| Virat Kohli                    |           0 |
| Virat Kohli                    |           8 |
| Virat Kohli                    |          64 |
| Virat Kohli                    |           0 |
| Virat Kohli                    |          37 |
| Virat Kohli                    |          37 |
| Virat Kohli                    |          17 |
| Virat Kohli                    |          52 |
| Virat Kohli                    |          42 |
| Virat Kohli                    |          14 |
| Virat Kohli                    |          55 |
| Virat Kohli                    |          46 |
| Virat Kohli                    |          40 |
| Virat Kohli                    |          39 |
| Virat Kohli                    |          35 |
| Virat Kohli                    |           0 |
| Virat Kohli                    |           1 |
| Virat Kohli                    |          12 |
| Virat Kohli                    |          69 |
| Virat Kohli                    |          79 |
| Virat Kohli                    |          36 |
| Virat Kohli                    |           4 |
| Virat Kohli                    |           6 |
| Virat Kohli                    |          30 |
| Virat Kohli                    |          53 |
```

Figure 6.12. Preserved SQL operation, analytics or result evidence from source report page 26.

```
mysql> SELECT p.player_name, bp.runs_scored
    -> FROM player p
    -> LEFT JOIN batting_performance bp
    -> ON p.player_id = bp.player_id;
+-------------------------------+-------------+
| player_name                   | runs_scored |
+-------------------------------+-------------+
| Virat Kohli                   |          43 |
| Virat Kohli                   |          12 |
| Virat Kohli                   |          53 |
| Virat Kohli                   |          20 |
| Virat Kohli                   |          36 |
| Virat Kohli                   |          18 |
| Virat Kohli                   |          42 |
| Virat Kohli                   |           8 |
| Virat Kohli                   |          84 |
| Virat Kohli                   |          68 |
| Virat Kohli                   |         172 |
| Virat Kohli                   |          16 |
| Virat Kohli                   |           3 |
| Virat Kohli                   |          47 |
| Virat Kohli                   |           7 |
| Virat Kohli                   |          27 |
| Virat Kohli                   |           0 |
| Virat Kohli                   |          28 |
| Virat Kohli                   |           0 |
| Virat Kohli                   |           3 |
| Virat Kohli                   |           1 |
| Virat Kohli                   |           1 |
| Virat Kohli                   |           0 |
| Virat Kohli                   |           8 |
| Virat Kohli                   |          64 |
| Virat Kohli                   |           0 |
| Virat Kohli                   |          37 |
| Virat Kohli                   |          37 |
| Virat Kohli                   |          17 |
| Virat Kohli                   |          52 |
| Virat Kohli                   |          42 |
| Virat Kohli                   |          14 |
| Virat Kohli                   |          55 |
| Virat Kohli                   |          46 |
| Virat Kohli                   |          40 |
| Virat Kohli                   |          39 |
| Virat Kohli                   |          35 |
| Virat Kohli                   |           0 |
| Virat Kohli                   |           1 |
| Virat Kohli                   |          12 |
| Virat Kohli                   |          69 |
| Virat Kohli                   |          79 |
| Virat Kohli                   |          36 |
| Virat Kohli                   |           4 |
| Virat Kohli                   |           6 |
| Virat Kohli                   |          30 |
| Virat Kohli                   |          53 |
| Virat Kohli                   |          34 |
| Virat Kohli                   |           8 |
| Virat Kohli                   |          25 |
| Virat Kohli                   |          35 |
| Virat Kohli                   |           2 |
| Virat Kohli                   |           8 |
```

Figure 6.13. Preserved SQL operation, analytics or result evidence from source report page 27.

```
mysql> SELECT p.player_name, bp.runs_scored
    -> FROM player p
    -> RIGHT JOIN batting_performance bp
    -> ON p.player_id = bp.player_id;
+--------------------------------+--------------+
| player_name                    | runs_scored  |
+--------------------------------+--------------+
| Virat Kohli                    |           43 |
| M Klinger                      |           38 |
| TM Head                        |           31 |
| MC Henriques                   |           17 |
| AJ Turner                      |           18 |
| JP Faulkner                    |           14 |
| N Dickwella                    |           30 |
| WU Tharanga                    |            0 |
| EMDY Munaweera                 |           44 |
| DAS Gunaratne                  |           52 |
| TAM Siriwardana                |           15 |
| CK Kapugedera                  |           10 |
| S Prasanna                     |            8 |
| M Klinger                      |           43 |
| Virat Kohli                    |           12 |
| BR Dunk                        |           32 |
| MC Henriques                   |           56 |
| TM Head                        |            4 |
| AJ Turner                      |            7 |
| JP Faulkner                    |            1 |
| TD Paine                       |            7 |
| PJ Cummins                     |            3 |
| AJ Tye                         |            0 |
| JA Richardson                  |            0 |
| N Dickwella                    |           14 |
| WU Tharanga                    |            4 |
| EMDY Munaweera                 |           10 |
| BKG Mendis                     |            5 |
| DAS Gunaratne                  |           84 |
| TAM Siriwardana                |            0 |
| CK Kapugedera                  |           32 |
| S Prasanna                     |            7 |
| KMDN Kulasekara                |           12 |
| SL Malinga                     |            1 |
| Virat Kohli                    |           53 |
| M Klinger                      |           62 |
| BR Dunk                        |           28 |
| TM Head                        |           30 |
| AJ Turner                      |            1 |
| MC Henriques                   |            2 |
| JP Faulkner                    |            2 |
| TD Paine                       |            1 |
| EMDY Munaweera                 |           37 |
| WU Tharanga                    |           14 |
| BKG Mendis                     |           14 |
```

Figure 6.14. Preserved SQL operation, analytics or result evidence from source report page 28.

```
mysql> SELECT p.player_name, bp.runs_scored
    -> FROM player p
    -> LEFT JOIN batting_performance bp
    -> ON p.player_id = bp.player_id
    ->
    -> UNION
    ->
    -> SELECT p.player_name, bp.runs_scored
    -> FROM player p
    -> RIGHT JOIN batting_performance bp
    -> ON p.player_id = bp.player_id;
+-----------------------------+-------------+
| player_name                 | runs_scored |
+-----------------------------+-------------+
| Virat Kohli                 |          43 |
| Virat Kohli                 |          12 |
| Virat Kohli                 |          53 |
| Virat Kohli                 |          20 |
| Virat Kohli                 |          36 |
| Virat Kohli                 |          18 |
| Virat Kohli                 |          42 |
| Virat Kohli                 |           8 |
| Virat Kohli                 |          84 |
| Virat Kohli                 |          68 |
| Virat Kohli                 |         172 |
| Virat Kohli                 |          16 |
| Virat Kohli                 |           3 |
| Virat Kohli                 |          47 |
| Virat Kohli                 |           7 |
| Virat Kohli                 |          27 |
| Virat Kohli                 |           0 |
| Virat Kohli                 |          28 |
| Virat Kohli                 |           1 |
| Virat Kohli                 |          64 |
| Virat Kohli                 |          37 |
| Virat Kohli                 |          17 |
| Virat Kohli                 |          52 |
| Virat Kohli                 |          14 |
| Virat Kohli                 |          55 |
| Virat Kohli                 |          46 |
| Virat Kohli                 |          40 |
| Virat Kohli                 |          39 |
| Virat Kohli                 |          35 |
| Virat Kohli                 |          69 |
| Virat Kohli                 |          79 |
| Virat Kohli                 |           4 |
| Virat Kohli                 |           6 |
| Virat Kohli                 |          30 |
| Virat Kohli                 |          34 |
| Virat Kohli                 |          25 |
| Virat Kohli                 |           2 |
| Virat Kohli                 |          44 |
| Virat Kohli                 |           9 |
| Virat Kohli                 |           5 |
| Virat Kohli                 |          13 |
| Virat Kohli                 |          31 |
| Virat Kohli                 |          63 |
| Virat Kohli                 |          61 |
| Virat Kohli                 |          24 |
| Virat Kohli                 |          29 |
| Virat Kohli                 |          58 |
```

Figure 6.15. Preserved SQL operation, analytics or result evidence from source report page 29.

```
mysql> SHOW FULL TABLES
    -> WHERE Table_type = 'VIEW';
+-------------------------------+------------+
| Tables_in_cricket_selection   | Table_type |
+-------------------------------+------------+
| vw_batting_performance        | VIEW       |
| vw_player_details             | VIEW       |
+-------------------------------+------------+
2 rows in set (0.00 sec)

mysql>
mysql> SELECT *
    -> FROM vw_player_details
    -> LIMIT 10;
+-----------+--------------+-----------+----------------+----------------------+---------------+
| player_id | player_name  | country   | batting_style  | bowling_style        | date_of_birth |
+-----------+--------------+-----------+----------------+----------------------+---------------+
|         1 | Virat Kohli  | India     | Right-handed   | NULL                 | NULL          |
|         2 | M Klinger    | Australia | Right hand Bat |                      | 1980-07-04    |
|         3 | TM Head      | Australia | Left hand Bat  | Right arm Offbreak   | 1993-12-29    |
|         4 | MC Henriques | Australia | Right hand Bat | Right arm Fast medium| 1987-02-01    |
|         5 | AJ Turner    | Australia | Right hand Bat | Right arm Offbreak   | 1993-01-25    |
|         6 | JP Faulkner  | Australia | Right hand Bat | Left arm Fast medium | 1990-04-29    |
|         7 | TD Paine     | Australia | Right hand Bat | Right arm Medium     | 1984-12-08    |
|         8 | PJ Cummins   | Australia | Right hand Bat | Right arm Fast       | 1993-05-08    |
|         9 | A Zampa      | Australia | Right hand Bat | Legbreak Googly      | 1992-03-31    |
|        10 | B Stanlake   | Australia | Left hand Bat  | Right arm Fast       | 1994-11-04    |
+-----------+--------------+-----------+----------------+----------------------+---------------+
10 rows in set (0.00 sec)
```

Figure 6.16. Preserved SQL operation, analytics or result evidence from source report page 30.

```
mysql>
mysql> SELECT *
    -> FROM vw_batting_performance
    -> LIMIT 10;
+-----------+--------------+-----------+------------+----------+--------------+------------+-------+-------+-------------+------------------+-----------+
| player_id | player_name  | country   | innings_id | match_id | runs_scored  | balls_faced| fours | sixes | strike_rate | batting_position | phase     |
+-----------+--------------+-----------+------------+----------+--------------+------------+-------+-------+-------------+------------------+-----------+
|         1 | Virat Kohli  | India     |          1 |        1 |           43 |         34 |     2 |     2 |      126.47 |                1 | Middle    |
|         2 | M Klinger    | Australia |          1 |        1 |           38 |         32 |     4 |     0 |      118.75 |                2 | Powerplay |
|         3 | TM Head      | Australia |          1 |        1 |           31 |         25 |     0 |     1 |      124.00 |                3 | Middle    |
|         4 | MC Henriques | Australia |          1 |        1 |           17 |         10 |     0 |     1 |      170.00 |                4 | Middle    |
|         5 | AJ Turner    | Australia |          1 |        1 |           18 |         12 |     1 |     0 |      150.00 |                5 | Middle    |
|         6 | JP Faulkner  | Australia |          1 |        1 |           14 |          7 |     2 |     0 |      200.00 |                6 | Death     |
|        12 | N Dickwella  | Sri Lanka |          2 |        1 |           30 |         25 |     4 |     1 |      120.00 |                1 | Powerplay |
|        13 | WU Tharanga  | Sri Lanka |          2 |        1 |            0 |          2 |     0 |     0 |        0.00 |                2 | Powerplay |
|        14 | EMDY Munaweera | Sri Lanka |          2 |        1 |           44 |         29 |     6 |     0 |      151.72 |                3 | Powerplay |
|        15 | DAS Gunaratne | Sri Lanka |          2 |        1 |           52 |         37 |     7 |     0 |      140.54 |                4 | Middle    |
+-----------+--------------+-----------+------------+----------+--------------+------------+-------+-------+-------------+------------------+-----------+
10 rows in set (0.00 sec)
```

Figure 6.17. Preserved SQL operation, analytics or result evidence from source report page 30.



```
mysql>
mysql> SHOW CREATE VIEW vw_player_details;
+------------------+-----------------------
```

```
mysql> SHOW PROCEDURE STATUS
    -> WHERE Db = 'cricket_selection';
+------------------+-----------------------+-----------+----------------+---------------------+---------------------+---------------+---------+---------
-------------+----------------------+--------------------+
| Db               | Name                  | Type      | Definer        | Modified            | Created             | Security_type | Comment | characte
r_set_client | collation_connection | Database Collation |
+------------------+-----------------------+-----------+----------------+---------------------+---------------------+---------------+---------+---------
-------------+----------------------+--------------------+
| cricket_selection | GetPlayerBattingStats | PROCEDURE | root@localhost | 2026-10-19 16:50:00 | 2026-10-19 16:50:00 | DEFINER       |         | cp850
            | cp850_general_ci     | utf8mb4_0900_ai_ci |
+------------------+-----------------------+-----------+----------------+---------------------+---------------------+---------------+---------+---------
-------------+----------------------+--------------------+
1 row in set (0.00 sec)

mysql>
mysql> SHOW CREATE PROCEDURE GetPlayerBattingStats;
+-----------------------+-----------------------
```



```
mysql> SHOW TRIGGERS;
+-------------------------------+--------+-------------------+-------------------------------
```

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> INSERT INTO player
    -> (player_name, country, batting_style)
    -> VALUES
    -> ('Transaction Demo Player', 'India', 'Right hand bat');
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> SELECT *
    -> FROM player
    -> WHERE player_name = 'Transaction Demo Player';
+-----------+--------------------------+---------+----------------+---------------+---------------+
| player_id | player_name              | country | batting_style  | bowling_style | date_of_birth |
+-----------+--------------------------+---------+----------------+---------------+---------------+
|      8017 | Transaction Demo Player | India   | Right hand bat | NULL          | NULL          |
+-----------+--------------------------+---------+----------------+---------------+---------------+
1 row in set (0.00 sec)

mysql>
mysql> UPDATE player
    -> SET country = 'Test Country'
    -> WHERE player_name = 'Transaction Demo Player';
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> SELECT *
    -> FROM player
    -> WHERE player_name = 'Transaction Demo Player';
+-----------+--------------------------+--------------+----------------+---------------+---------------+
| player_id | player_name              | country      | batting_style  | bowling_style | date_of_birth |
+-----------+--------------------------+--------------+----------------+---------------+---------------+
|      8017 | Transaction Demo Player | Test Country | Right hand bat | NULL          | NULL          |
+-----------+--------------------------+--------------+----------------+---------------+---------------+
1 row in set (0.01 sec)

mysql>
mysql> ROLLBACK;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> SELECT *
    -> FROM player
    -> WHERE player_name = 'Transaction Demo Player';
Empty set (0.01 sec)
```

Figure 6.23. Preserved SQL operation, analytics or result evidence from source report page 33.

```
mysql>
mysql> SELECT *
    -> FROM player
    -> WHERE player_name = 'Transaction Demo Player';
Empty set (0.01 sec)

mysql> WITH PlayerTotals AS (
    ->     SELECT
    ->         p.player_id,
    ->         p.player_name,
    ->         p.country,
    ->         SUM(bp.runs_scored) AS total_runs
    ->     FROM player p
    ->     JOIN batting_performance bp
    ->     ON p.player_id = bp.player_id
    ->     GROUP BY p.player_id, p.player_name, p.country
    -> )
    -> SELECT *
    -> FROM PlayerTotals;
+-----------+------------------------------+-------------------------+------------+
| player_id | player_name                  | country                 | total_runs |
+-----------+------------------------------+-------------------------+------------+
|         1 | Virat Kohli                  | India                   |       8308 |
|         2 | M Klinger                    | Australia               |        143 |
|         3 | TM Head                      | Australia               |       4283 |
|         4 | MC Henriques                 | Australia               |        650 |
|         5 | AJ Turner                    | Australia               |        302 |
|         6 | JP Faulkner                  | Australia               |       1184 |
|        12 | N Dickwella                  | Sri Lanka               |       2084 |
|        13 | WU Tharanga                  | Sri Lanka               |       6889 |
|        14 | EMDY Munaweera               | Sri Lanka               |        230 |
|        15 | DAS Gunaratne                | Sri Lanka               |        800 |
|        16 | TAM Siriwardana              | Sri Lanka               |        791 |
|        17 | CK Kapugedera                | Sri Lanka               |       2180 |
|        18 | S Prasanna                   | Sri Lanka               |        635 |
|        23 | BR Dunk                      | Australia               |         99 |
|         7 | TD Paine                     | Australia               |        943 |
|         8 | PJ Cummins                   | Australia               |        680 |
|        11 | AJ Tye                       | Australia               |        140 |
|        24 | JA Richardson                | Australia               |        138 |
|        25 | BKG Mendis                   | Sri Lanka               |       7001 |
|        21 | KMDN Kulasekara              | Sri Lanka               |       1519 |
|        20 | SL Malinga                   | Sri Lanka               |        644 |
|        26 | MD Shanaka                   | Sri Lanka               |       3133 |
|        19 | JRMVB Sanjaya                | Sri Lanka               |         20 |
|        27 | Nizakat Khan                 | Hong Kong               |       2896 |
|        28 | Aizaz Khan                   | Hong Kong               |       1218 |
|        29 | Babar Hayat                  | Hong Kong               |       2954 |
|        30 | Anshuman Rath                | Hong Kong               |       2700 |
|        31 | Ehsan Khan                   | Hong Kong               |        476 |
|        32 | Tanwir Afzal                 | Hong Kong               |        317 |
|        33 | Shahid Wasif                 | Hong Kong               |        612 |
|        38 | WTS Porterfield              | Ireland                 |       3390 |
|        39 | DA Rankin                    | Ireland                 |         15 |
|        40 | SP Terry                     | Ireland                 |         20 |
|        41 | JNK Shannon                  | Ireland                 |        111 |
|        43 | GJ Thompson                  | Ireland                 |         93 |
|        42 | KJ O'Brien                   | Ireland                 |       3561 |
|        44 | AR McBrine                   | Ireland                 |        946 |
|        45 | L Tucker                     | Ireland                 |       2420 |
|        46 | J Mulder                     | Ireland                 |          1 |
|        47 | J Little                     | Ireland                 |        204 |
|        48 | CA Young                     | Ireland                 |        222 |
|        60 | CJ Chibhabha                 | Zimbabwe                |       2499 |
|        61 | H Masakadza                  | Zimbabwe                |       5185 |
|        62 | R Mutumbami                  | Zimbabwe                |        583 |
|        63 | Sikandar Raza                | Zimbabwe                |       6392 |
|        64 | MN Waller                    | Zimbabwe                |       1546 |
|        65 | CT Mutombodzi                | Zimbabwe                |        235 |
|        66 | E Chigumbura                 | Zimbabwe                |       4035 |
|        67 | AG Cremer                    | Zimbabwe                |        572 |
|        68 | N Madziva                    | Zimbabwe                |        121 |
|        49 | KL Rahul                     | India                   |       5432 |
|        51 | AT Rayudu                    | India                   |       1679 |
|        50 | Mandeep Singh                | India                   |        146 |
```

Figure 6.24. Preserved SQL operation, analytics or result evidence from source report page 34.

7. NoSQL / Vector Database Commands

MongoDB database cricket_selection_nosql stores scouting reports in the scouting_reports collection. The source report demonstrates database selection, collection inspection and document operations. The application supports report CRUD and an aggregation that groups records by recommendation. The supplied materials do not establish a vector database or semantic vector search implementation.

Database Selection and Collection Inspection

```
use cricket_selection_nosql
show collections
db.scouting_reports.find().limit(5).pretty()
```

Document Operations and Aggregation

The project reports insertion, retrieval, update and deletion operations on scouting reports, alongside an aggregation demonstration. The original mongosh screenshots are retained as implementation evidence.

```
mysql> WITH PlayerTotals AS (
    ->     SELECT
    ->         p.player_id,
    ->         p.player_name,
    ->         SUM(bp.runs_scored) AS total_runs
    ->     FROM player p
    ->     JOIN batting_performance bp
    ->     ON p.player_id = bp.player_id
    ->     GROUP BY p.player_id, p.player_name
    -> )
    -> SELECT
    ->     player_name,
    ->     total_runs,
    ->     RANK() OVER (
    ->         ORDER BY total_runs DESC
    ->     ) AS player_rank
    -> FROM PlayerTotals
    -> ORDER BY player_rank;
+------------------------------+------------+-------------+
| player_name                  | total_runs | player_rank |
+------------------------------+------------+-------------+
| V Kohli                      |      18788 |           1 |
| RG Sharma                    |      15574 |           2 |
| KC Sangakkara                |      13022 |           3 |
| MS Dhoni                     |      11858 |           4 |
| AB de Villiers               |      11026 |           5 |
| TM Dilshan                   |      10959 |           6 |
| Babar Azam                   |      10873 |           7 |
| MJ Guptill                   |      10642 |           8 |
| LRPL Taylor                  |      10100 |           9 |
| Q de Kock                    |      10000 |          10 |
| DA Warner                    |       9877 |          11 |
| JC Buttler                   |       9710 |          12 |
| KS Williamson                |       9682 |          13 |
| DPMD Jayawardene             |       9531 |          14 |
| EJG Morgan                   |       9396 |          15 |
| Tamim Iqbal                  |       9272 |          16 |
| HM Amla                      |       9106 |          17 |
| S Mandhana                   |       8968 |          18 |
| Shakib Al Hasan              |       8933 |          19 |
| Mohammad Hafeez              |       8788 |          20 |
| S Dhawan                     |       8492 |          21 |
| JE Root                      |       8459 |          22 |
| Mushfiqur Rahim              |       8403 |          23 |
| Shoaib Malik                 |       8373 |          24 |
| CH Gayle                     |       8365 |          25 |
| Virat Kohli                  |       8308 |          26 |
| L Wolvaardt                  |       8217 |          27 |
| Yuvraj Singh                 |       8068 |          28 |
| SD Hope                      |       7531 |          29 |
| MJ Clarke                    |       7476 |          30 |
| RT Ponting                   |       7380 |          31 |
| Mahmudullah                  |       7354 |          32 |
| DA Miller                    |       7351 |          33 |
| BB McCullum                  |       7302 |          34 |
| SK Raina                     |       7134 |          35 |
| BKG Mendis                   |       7001 |          36 |
| AD Mathews                   |       6989 |          37 |
| F du Plessis                 |       6932 |          38 |
| BL Mooney                    |       6894 |          39 |
```

Figure 7.1. Preserved MongoDB command or aggregation evidence from source report page 35.



Figure 7.2. Preserved MongoDB command or aggregation evidence from source report page 36.

```
> db.scouting_reports.countDocuments()
< 3
> db.scouting_reports.findOne()
< {
    _id: ObjectId('6ae5a987be17bc508e85a1c4'),
    playerId: 1,
    playerName: 'Virat Kohli',
    recommendation: 'Highly Recommended',
    role: 'Batsman',
    strengths: [
      'Consistency',
      'Chasing ability',
      'Experience'
    ],
    notes: 'Strong batting performance and match experience.'
  }
```

Figure 7.3. Preserved MongoDB command or aggregation evidence from source report page 36.

```
> db.scouting_reports.find().limit(5)
< {
    _id: ObjectId('6ae5a987be17bc508e85a1c4'),
    playerId: 1,
    playerName: 'Virat Kohli',
    recommendation: 'Highly Recommended',
    role: 'Batsman',
    strengths: [
      'Consistency',
      'Chasing ability',
      'Experience'
    ],
    notes: 'Strong batting performance and match experience.'
  }
  {
    _id: ObjectId('6ae5a987be17bc508e85a1c5'),
    playerId: 4405,
    playerName: 'A Adekunle',
    recommendation: 'Recommended',
    role: 'Batsman',
    strengths: [
      'Batting technique',
      'Consistency'
    ],
    notes: 'Promising player with good batting potential.'
  }
```

Figure 7.4. Preserved MongoDB command or aggregation evidence from source report page 37.



Figure 7.5. Preserved MongoDB command or aggregation evidence from source report page 37.

8. Results

The source report provides MySQL table and query screenshots, MongoDB command evidence, and web application demonstrations. Together they document database structures, analytics outputs, scouting operations and application behavior. The selected result figures below reuse original project screenshots.

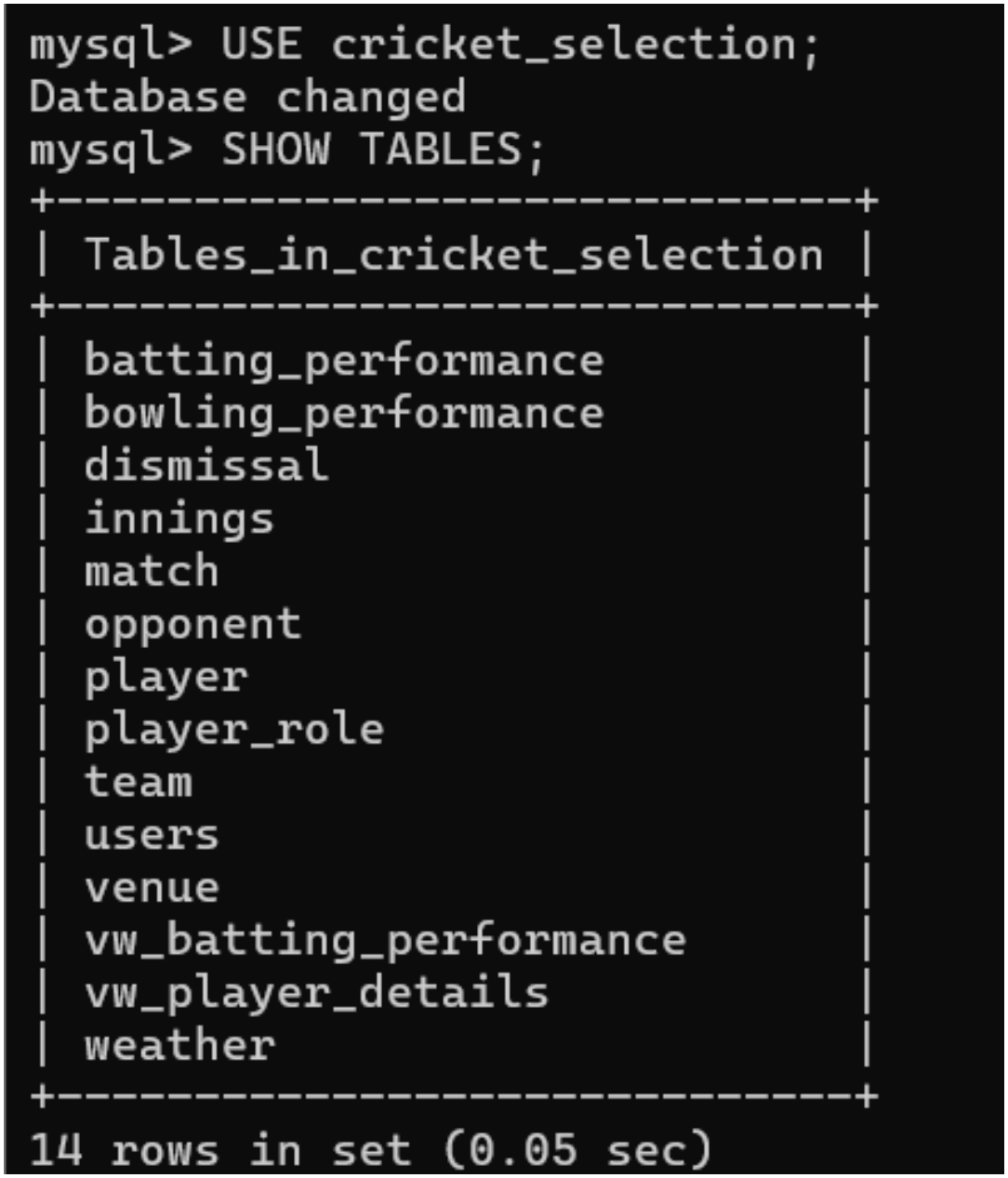


Figure 8.1. MySQL table structure evidence.

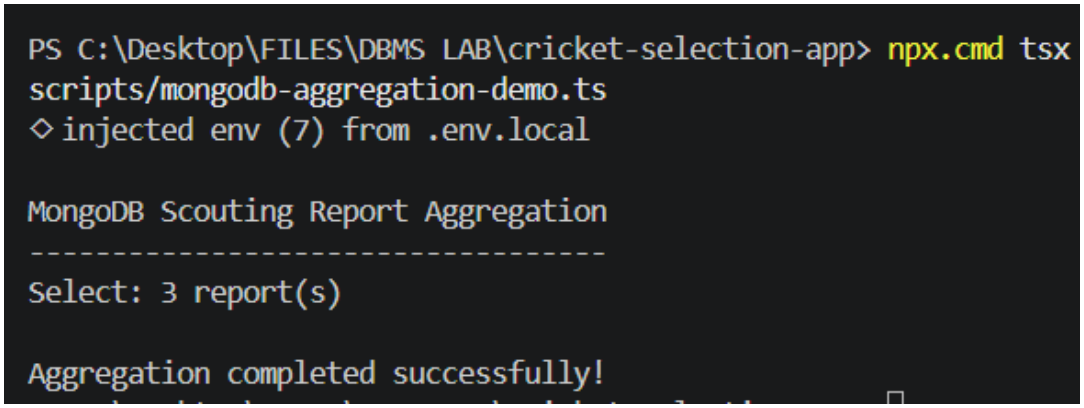


Figure 8.2. MongoDB scouting-report aggregation evidence.

Result Interpretation

The MySQL model organizes cricket data across 12 tables and provides two views for reusable player and batting-performance queries. SQL reports support comparative performance analysis. MongoDB stores flexible scouting information and groups reports by recommendation. The web interface presents player, selection, analytics and scouting workflows. The figures are evidence from the supplied report and project folder; this document does not infer unshown query values.

9. Next.js – Project Tree Structure

The following tree lists the main application, API, database connection, authentication, SQL, scripts and documentation paths present in the project directory. Build output, dependencies and environment files are excluded.

```
cricket-selection-app/
├── app/
│   ├── page.tsx
│   ├── PlayerSelection.tsx
│   ├── login/page.tsx
│   ├── admin/users/page.tsx
│   ├── scouting-reports/page.tsx
│   └── api/
│       ├── admin/users/route.ts
│       ├── advanced-reports/route.ts
│       ├── auth/login/route.ts
│       ├── auth/logout/route.ts
│       ├── players/route.ts
│       ├── reports/route.ts
│       ├── scouting-reports/route.ts
│       └── selection/route.ts
├── lib/
│   ├── auth.ts
```

```
│   ├── db.ts
│   ├── mongodb.ts
│   └── requireAdmin.ts
├── sql/
│   ├── sql-operations-demo.sql
│   └── advanced/{views,stored-procedure,trigger}.sql
├── scripts/
│   ├── create-users.ts
│   ├── mongodb-aggregation-demo.ts
│   └── test-mongodb.ts
├── docs/
│   ├── database-design.md
│   ├── data-dictionary.md
│   ├── database-technology.md
│   ├── ER_Diagram.md
│   └── screenshots/
├── proxy.ts
├── next.config.ts
└── package.json
```

Database connection settings are read through environment variables. The report does not reproduce environment-file contents or credentials.

10. Application Code and Demo Screenshots

The original PDF's application and demonstration images are preserved here with captions. They show the application interface and project evidence; the exact screenshots are reproduced without replacing them with recreated images.

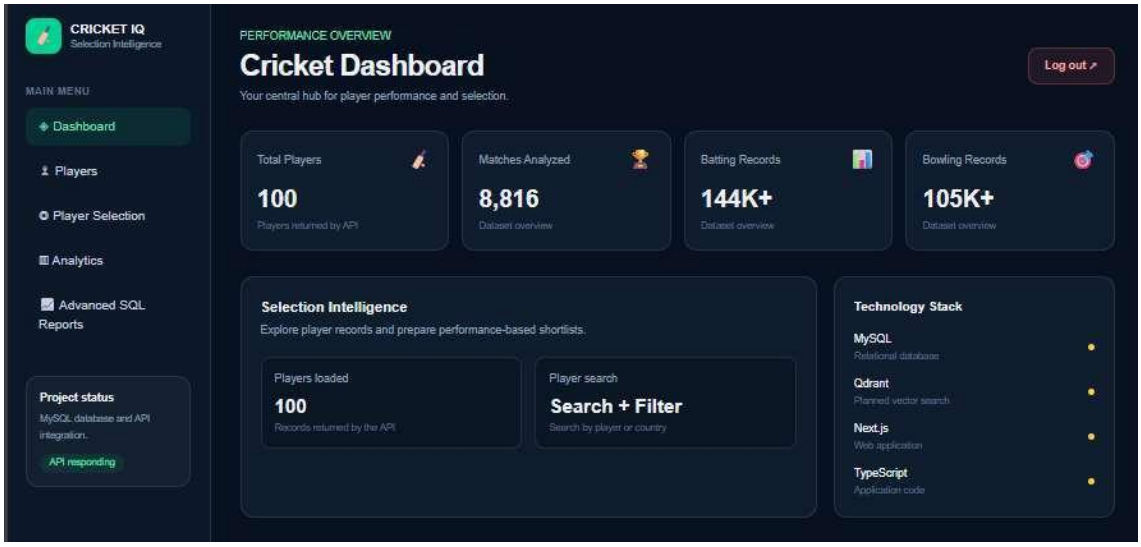
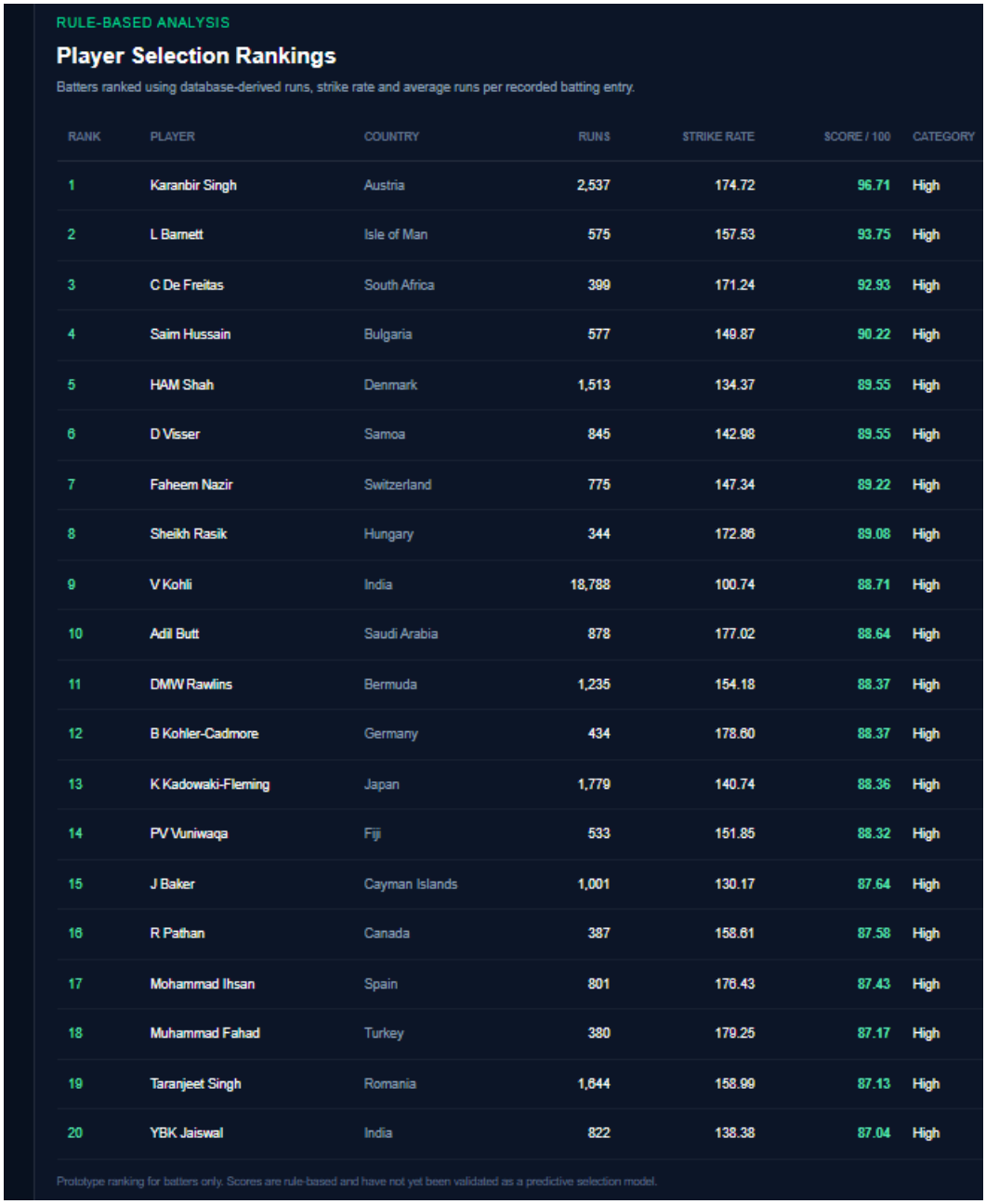


Figure 10.1. Application code or demo screenshot preserved from source report page 38.



RULE-BASED ANALYSIS

Player Selection Rankings

Batters ranked using database-derived runs, strike rate and average runs per recorded batting entry.

RANK	PLAYER	COUNTRY	RUNS	STRIKE RATE	SCORE / 100	CATEGORY
1	Karanbir Singh	Austria	2,537	174.72	96.71	High
2	L Barnett	Isle of Man	575	157.53	93.75	High
3	C De Freitas	South Africa	399	171.24	92.93	High
4	Saim Hussain	Bulgaria	577	149.87	90.22	High
5	HAM Shah	Denmark	1,513	134.37	89.55	High
6	D Visser	Samoa	845	142.98	89.55	High
7	Faheem Nazir	Switzerland	775	147.34	89.22	High
8	Sheikh Rasik	Hungary	344	172.86	89.08	High
9	V Kohli	India	18,788	100.74	88.71	High
10	Adil Butt	Saudi Arabia	878	177.02	88.64	High
11	DMW Rawlins	Bermuda	1,235	154.18	88.37	High
12	B Kohler-Cadmore	Germany	434	178.60	88.37	High
13	K Kadowaki-Fleming	Japan	1,779	140.74	88.36	High
14	PV Vuniwaqa	Fiji	533	151.85	88.32	High
15	J Baker	Cayman Islands	1,001	130.17	87.64	High
16	R Pathan	Canada	387	158.61	87.58	High
17	Mohammad Ihsan	Spain	801	176.43	87.43	High
18	Muhammad Fahad	Turkey	380	179.25	87.17	High
19	Taranjeet Singh	Romania	1,644	158.99	87.13	High
20	YBK Jaiswal	India	822	138.38	87.04	High

Prototype ranking for batters only. Scores are rule-based and have not yet been validated as a predictive selection model.

Figure 10.2. Application code or demo screenshot preserved from source report page 39.

SQL ANALYTICS

Top Batters Report

Ranked by total runs recorded in the dataset.

RANK	PLAYER	COUNTRY	RUNS	RECORDS	FOURS	SIXES	RUNS/RECORD
1	V Kohli	India	18,788	412	1,728	288	45.60
2	RG Sharma	India	15,574	418	1,452	553	37.26
3	KC Sangakkara	Sri Lanka	13,022	337	1,261	97	38.64
4	MS Dhoni	India	11,858	361	901	259	32.85
5	AB de Villiers	South Africa	11,026	285	960	254	38.69
6	TM Dilshan	Sri Lanka	10,959	335	1,232	79	32.71
7	Babar Azam	Pakistan	10,873	268	1,060	146	40.57
8	MJ Guptill	New Zealand	10,642	312	1,041	357	34.11
9	LRPL Taylor	New Zealand	10,100	307	800	207	32.90
10	Q de Kock	South Africa	10,000	263	1,092	259	38.02
11	DA Warner	Australia	9,877	264	1,034	247	37.41
12	JC Buttler	England	9,710	316	833	372	30.73
13	KS Williamson	New Zealand	9,682	254	899	119	38.12
14	DPMD Jayawardene	Sri Lanka	9,531	317	911	85	30.07
15	EJG Morgan	England	9,396	321	785	310	29.27
16	Tamim Iqbal	Bangladesh	9,272	286	1,013	141	32.42
17	HM Amla	South Africa	9,106	217	941	75	41.96
18	S Mandhana	India	8,968	248	1,151	154	36.16
19	Shakib Al Hasan	Bangladesh	8,933	310	854	100	28.82
20	Mohammad Hafeez	Pakistan	8,788	307	881	180	28.63

Figure 10.3. Application code or demo screenshot preserved from source report page 40.



ADVANCED SQL

Advanced SQL Reports

Top batters ranked using a CTE, JOIN, aggregation, and the RANK window function.

RANK	PLAYER	COUNTRY	TOTAL RUNS
1	V Kohli	India	18,788
2	RG Sharma	India	15,574
3	KC Sangakkara	Sri Lanka	13,022
4	MS Dhoni	India	11,858
5	AB de Villiers	South Africa	11,026
6	TM Dilshan	Sri Lanka	10,959
7	Babar Azam	Pakistan	10,873
8	MJ Guptill	New Zealand	10,642
9	LRPL Taylor	New Zealand	10,100
10	Q de Kock	South Africa	10,000
11	DA Warner	Australia	9,877
12	JC Buttler	England	9,710
13	KS Williamson	New Zealand	9,682
14	DPMD Jayawardene	Sri Lanka	9,531
15	EJG Morgan	England	9,396
16	Tamim Iqbal	Bangladesh	9,272
17	HM Amla	South Africa	9,106
18	S Mandhana	India	8,968
19	Shakib Al Hasan	Bangladesh	8,933
20	Mohammad Hafeez	Pakistan	8,788

Figure 10.4. Application code or demo screenshot preserved from source report page 41.

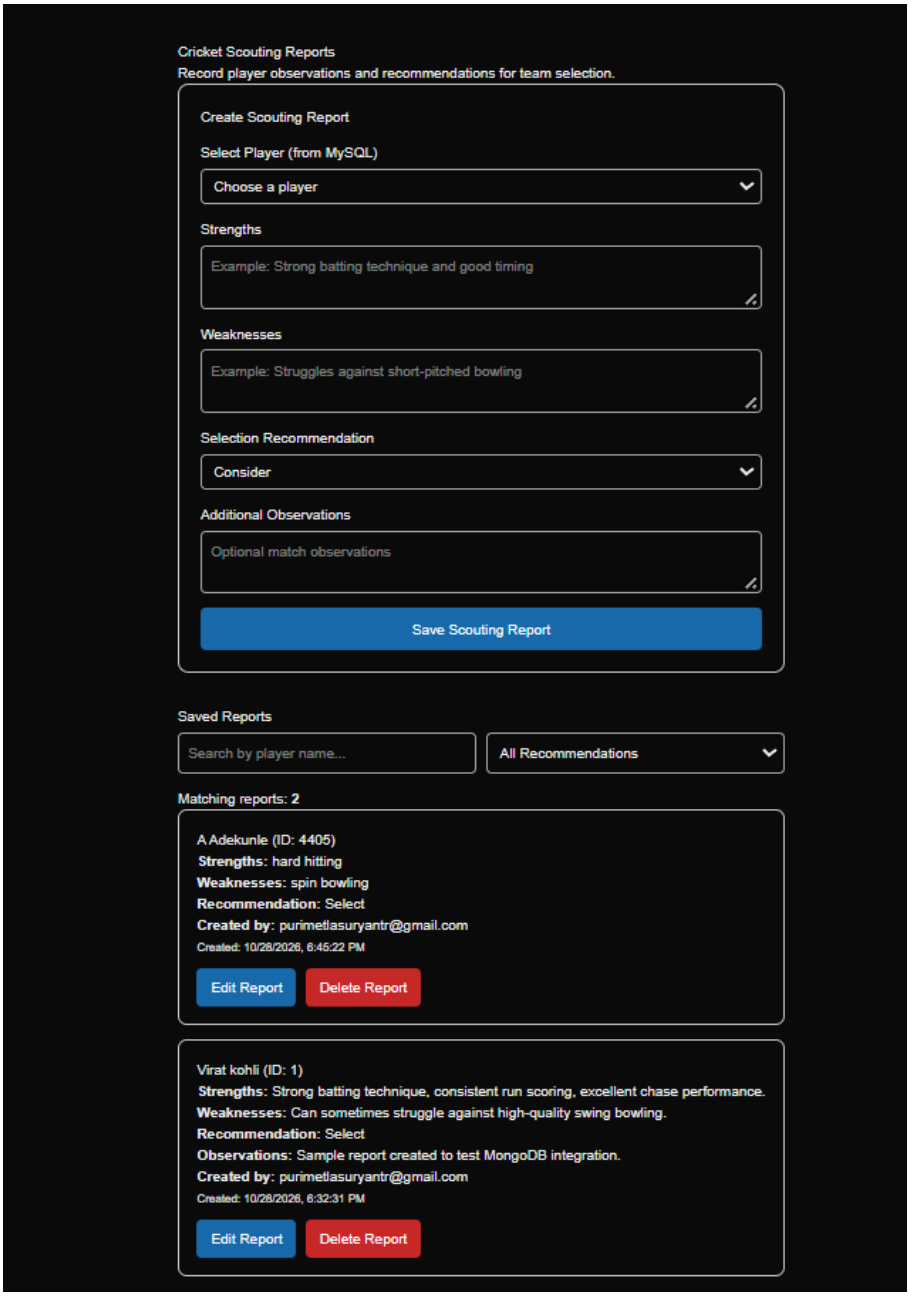


Figure 10.5. Application code or demo screenshot preserved from source report page 42.

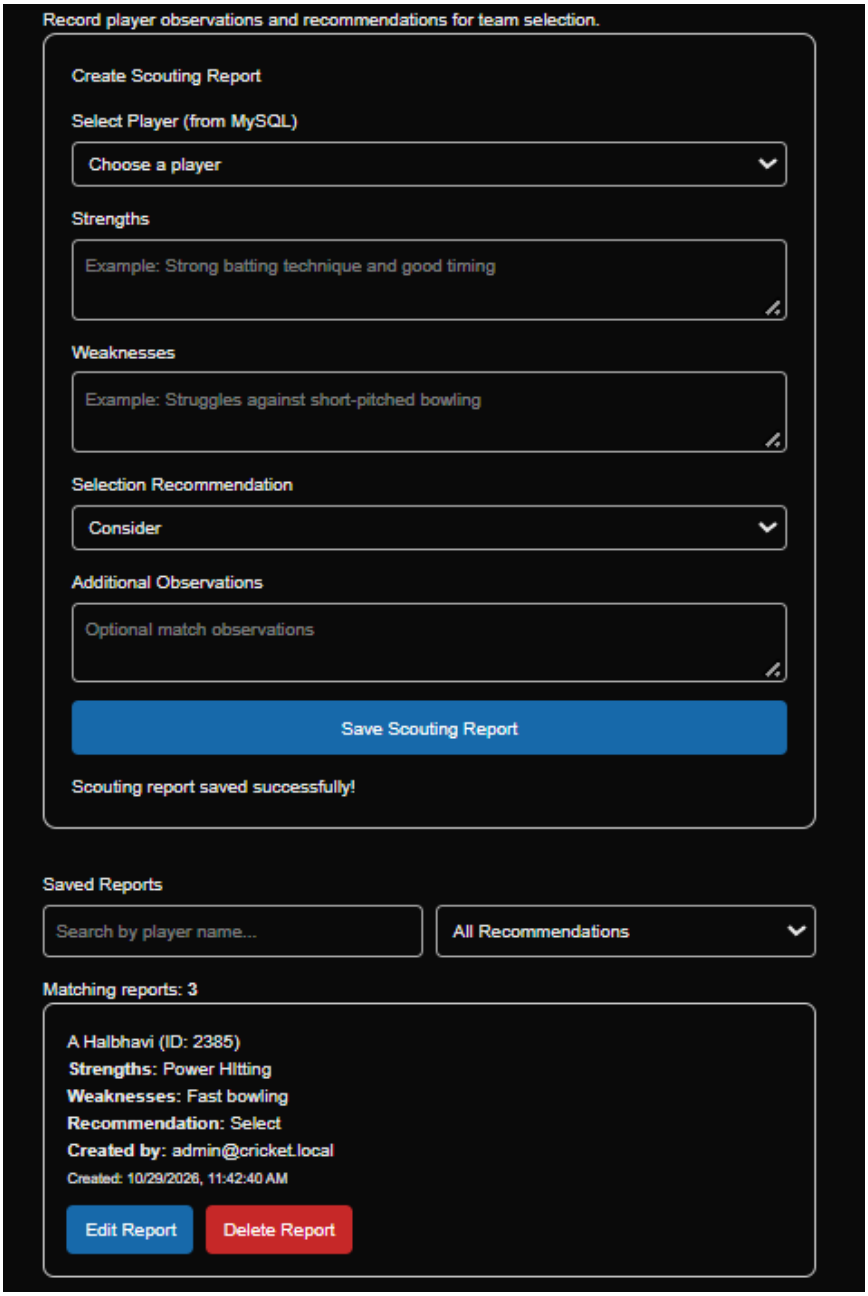


Figure 10.6. Application code or demo screenshot preserved from source report page 43.

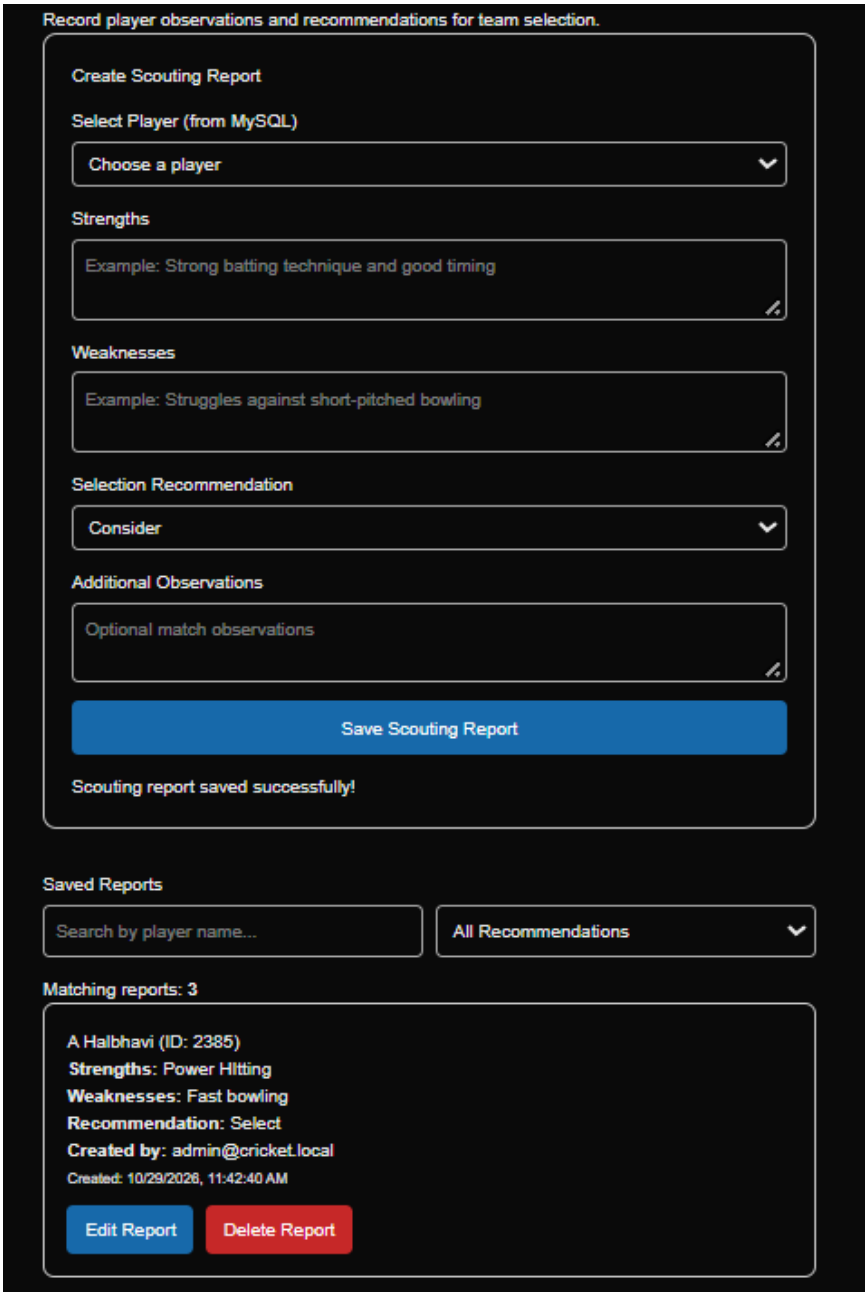


Figure 10.7. Application code or demo screenshot preserved from source report page 44.

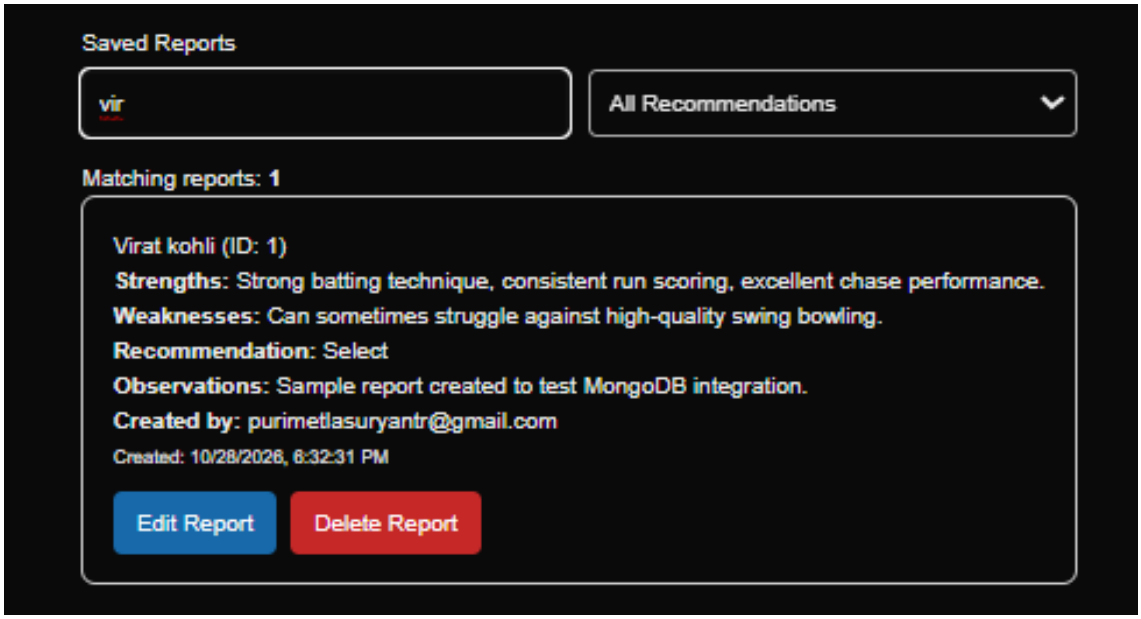


Figure 10.8. Application code or demo screenshot preserved from source report page 44.

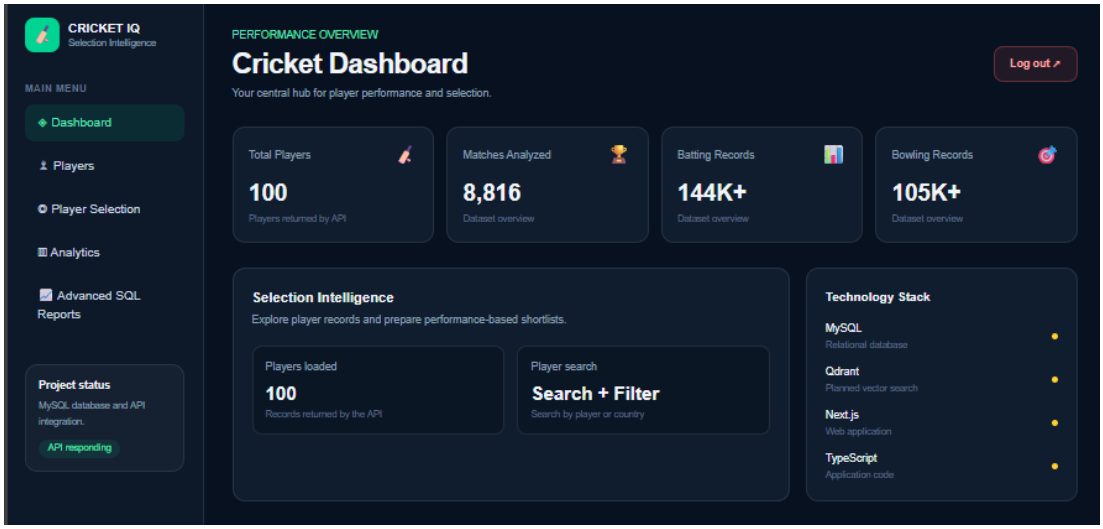


Figure 10.9. Application dashboard.

RULE-BASED ANALYSIS

Player Selection Rankings

Batters ranked using database-derived runs, strike rate and average runs per recorded batting entry.

RANK	PLAYER	COUNTRY	RUNS	STRIKE RATE	SCORE / 100	CATEGORY
1	Karanbir Singh	Austria	2,537	174.72	96.71	High
2	L Barnett	Isle of Man	575	157.53	93.75	High
3	C De Freitas	South Africa	399	171.24	92.93	High
4	Saim Hussain	Bulgaria	577	149.87	90.22	High
5	HAM Shah	Denmark	1,513	134.37	89.55	High
6	D Visser	Samoa	845	142.98	89.55	High
7	Faheem Nazir	Switzerland	775	147.34	89.22	High
8	Sheikh Rasik	Hungary	344	172.86	89.08	High
9	V Kohli	India	18,788	100.74	88.71	High
10	Adil Butt	Saudi Arabia	878	177.02	88.64	High
11	DMW Rawlins	Bermuda	1,235	154.18	88.37	High
12	B Kohler-Cadmore	Germany	434	178.60	88.37	High
13	K Kadowaki-Fleming	Japan	1,779	140.74	88.36	High
14	PV Vuniwaqa	Fiji	533	151.85	88.32	High
15	J Baker	Cayman Islands	1,001	130.17	87.64	High
16	R Pathan	Canada	387	158.61	87.58	High
17	Mohammad Ihsan	Spain	801	176.43	87.43	High
18	Muhammad Fahad	Turkey	380	179.25	87.17	High
19	Taranjeet Singh	Romania	1,644	158.99	87.13	High
20	YBK Jaiswal	India	822	138.38	87.04	High

Prototype ranking for batters only. Scores are rule-based and have not yet been validated as a predictive selection model.

Figure 10.10. Player selection.

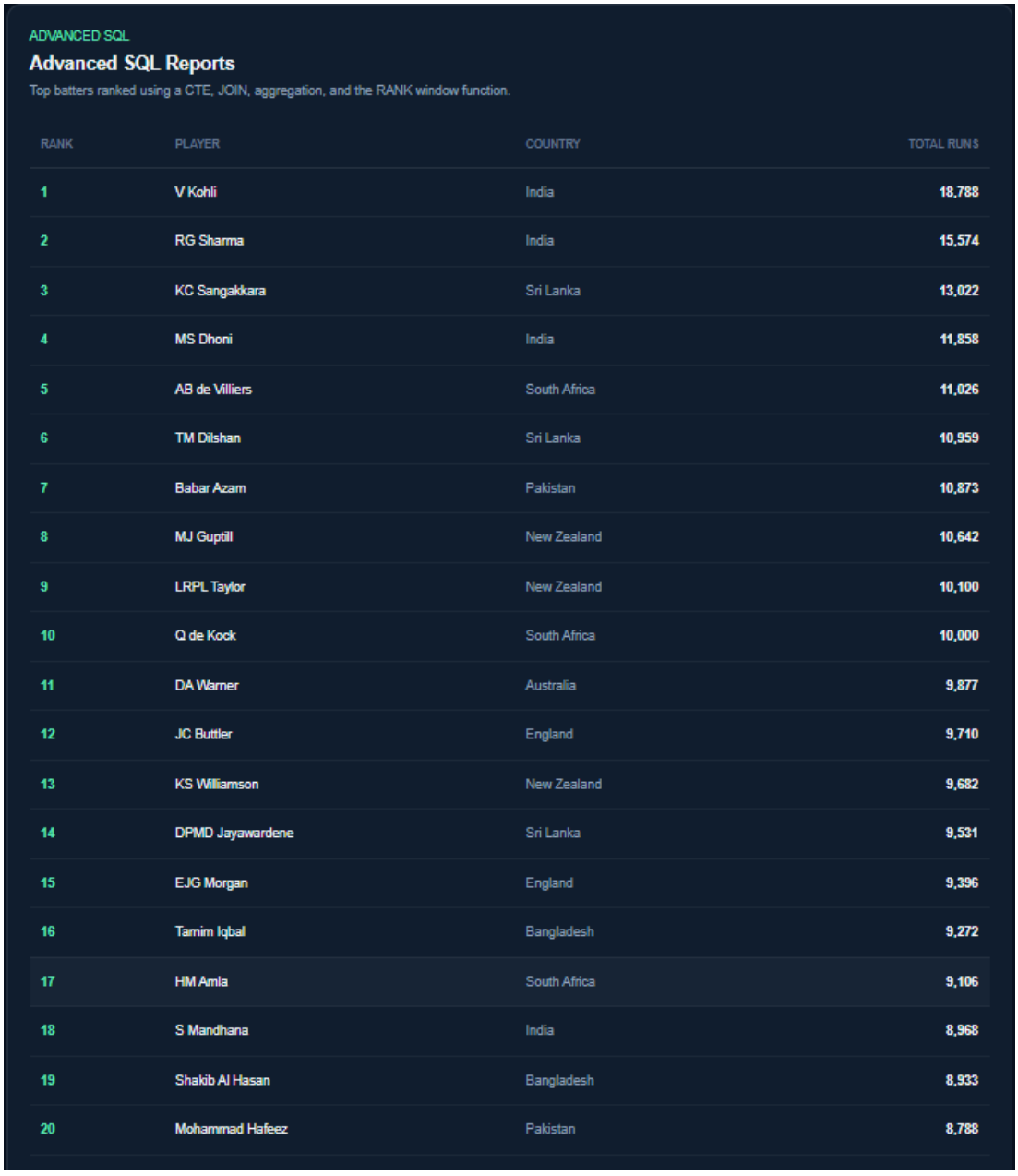
SQL ANALYTICS

Top Batters Report

Ranked by total runs recorded in the dataset.

RANK	PLAYER	COUNTRY	RUNS	RECORDS	FOURS	SIXES	RUNS/RECORD
1	V Kohli	India	18,788	412	1,728	288	45.60
2	RG Sharma	India	15,574	418	1,452	553	37.26
3	KC Sangakkara	Sri Lanka	13,022	337	1,261	97	38.64
4	MS Dhoni	India	11,858	361	901	259	32.85
5	AB de Villiers	South Africa	11,026	285	960	254	38.69
6	TM Dilshan	Sri Lanka	10,959	335	1,232	79	32.71
7	Babar Azam	Pakistan	10,873	268	1,060	146	40.57
8	MJ Guptill	New Zealand	10,642	312	1,041	357	34.11
9	LRPL Taylor	New Zealand	10,100	307	800	207	32.90
10	Q de Kock	South Africa	10,000	263	1,092	259	38.02
11	DA Warner	Australia	9,877	264	1,034	247	37.41
12	JC Buttler	England	9,710	316	833	372	30.73
13	KS Williamson	New Zealand	9,682	254	899	119	38.12
14	DPMD Jayawardene	Sri Lanka	9,531	317	911	85	30.07
15	EJG Morgan	England	9,396	321	785	310	29.27
16	Tamim Iqbal	Bangladesh	9,272	286	1,013	141	32.42
17	HM Amla	South Africa	9,106	217	941	75	41.96
18	S Mandhana	India	8,968	248	1,151	154	36.16
19	Shakib Al Hasan	Bangladesh	8,933	310	854	100	28.82
20	Mohammad Hafeez	Pakistan	8,788	307	881	180	28.63

Figure 10.11. Player performance analytics.



RANK	PLAYER	COUNTRY	TOTAL RUNS
1	V Kohli	India	18,788
2	RG Sharma	India	15,574
3	KC Sangakkara	Sri Lanka	13,022
4	MS Dhoni	India	11,858
5	AB de Villiers	South Africa	11,026
6	TM Dilshan	Sri Lanka	10,959
7	Babar Azam	Pakistan	10,873
8	MJ Guptill	New Zealand	10,642
9	LRPL Taylor	New Zealand	10,100
10	Q de Kock	South Africa	10,000
11	DA Warner	Australia	9,877
12	JC Buttler	England	9,710
13	KS Williamson	New Zealand	9,682
14	DPMD Jayawardene	Sri Lanka	9,531
15	EJG Morgan	England	9,396
16	Tamim Iqbal	Bangladesh	9,272
17	HM Amla	South Africa	9,106
18	S Mandhana	India	8,968
19	Shakib Al Hasan	Bangladesh	8,933
20	Mohammad Hafeez	Pakistan	8,788

Figure 10.12. Advanced SQL reports.

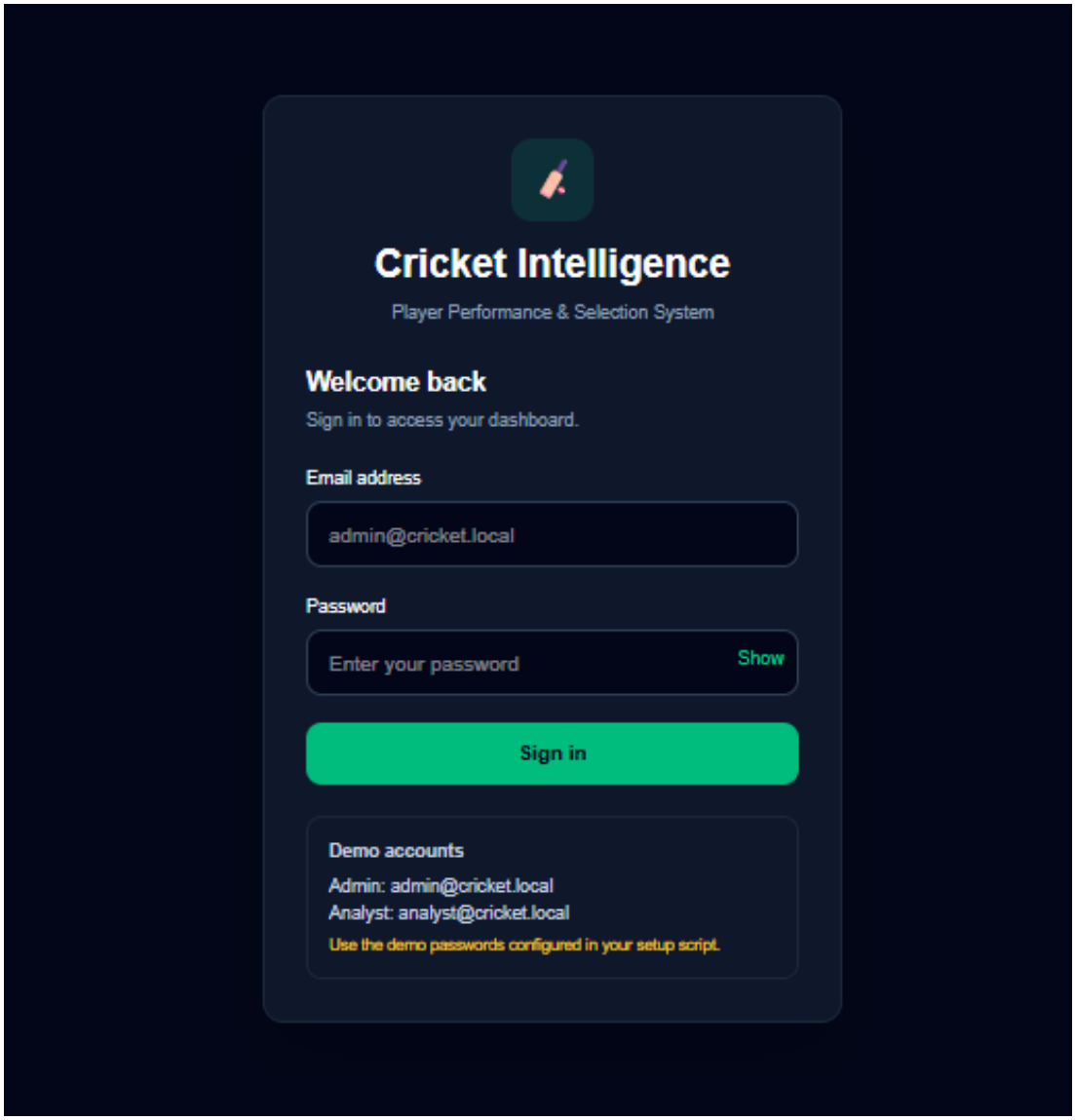


Figure 10.13. Authentication page.

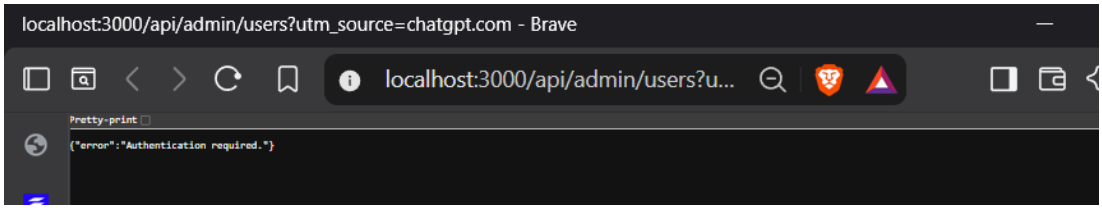


Figure 10.14. Role-based access demonstration.

```
test> show dbs
admin                   40.00 KiB
config                  96.00 KiB
cricket_selection_nosql  64.00 KiB
local                   40.00 KiB
test> use cricket_selection
switched to db cricket_selection
cricket_selection> show collections

cricket_selection> use cricket_selection_nosql
switched to db cricket_selection_nosql
cricket_selection_nosql> show collections
scouting_reports
cricket_selection_nosql> db.scouting_reports.find().limit(5).pretty()
[
  {
    _id: ObjectId('6ae1f267571a52c26ed19382'),
    playerId: 1,
    playerName: 'Virat kohli',
    strengths: 'Strong batting technique, consistent run scoring, excellent chase performance.',
    weaknesses: 'Can sometimes struggle against high-quality swing bowling.',
    recommendation: 'Select',
    observations: 'Sample report created to test MongoDB integration.',
    createdBy: 'purimetlasuryantr@gmail.com',
    createdAt: ISODate('2026-10-28T13:02:31.322Z')
  },
  {
    _id: ObjectId('6ae1f56a571a52c26ed19383'),
    playerId: 4405,
    playerName: 'A Adekunle',
    strengths: 'hard hitting',
    weaknesses: 'spin bowling',
    recommendation: 'Select',
    observations: '',
    createdBy: 'purimetlasuryantr@gmail.com',
    createdAt: ISODate('2026-10-28T13:15:22.333Z')
  },
  {
    _id: ObjectId('6ae2e3d8d56049dceea87e89'),
    playerId: 2385,
    playerName: 'A Halbhavi',
    strengths: 'Power HItting',
    weaknesses: 'Fast bowling',
    recommendation: 'Select',
    observations: '',
    createdBy: 'admin@cricket.local',
    createdAt: ISODate('2026-10-29T06:12:40.892Z')
  }
]
```

Figure 10.15. MongoDB scouting reports.

11. Evaluation Rubrics – Completion Status

The rubric categories below follow the official template. The status cells are left for faculty evaluation.

#	Criteria	Distribution	Status
1	Database Design	ER diagram; relational schema with keys, relation-ships and constraints	

2	Database Creation	SQL implementation; NoSQL database creation; technology justification	
3	SQL Operations and Queries	DML; nested queries; joins and analytics	
4	Advanced Database Concepts	Views; procedure; trigger; indexes; MongoDB operations and aggregation	
5	Web Application and Database Integration	Next.js interface; backend/API and database integration	
6	Authentication and Authorization	Authentication, admin and analyst roles, protected API operations	
7	Documentation, Git/GitHub, Demonstration, Testing, and Originality	Documentation, source control, demonstration, testing and contributions	